Continuity and Limits

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§1 Introduction

When we talk about the continuity of a function, the first thing that comes to our mind is the graph of a function that leaves no space or gap in-between the points on the graph, i.e., a graph which can be drawn without lifting a pen. While it's tempting to think of a continuous function simply as one whose graph you can draw without lifting your pen, this "no pen-lifting" idea – really path-connectedness – is stronger than what continuity formally requires. We shall learn the definitions of continuity in the following section.

§2 Definitions of continuity

Definition 2.1 (Sequential definition of continuity)

Let $f : D \to \mathbb{R}$ be a function. We say f is *continuous* at $a \in D$, if for every sequence x_n in D such that $x_n \to a$, it holds that $f(x_n) \to f(a)$.

Definition 2.2 ($\varepsilon - \delta$ definition of continuity)

Let $f : D \to \mathbb{R}$ be a function. We say f is *continuous* at $a \in D$ if for any $\varepsilon > 0$, $\exists\ \delta_\varepsilon > 0$ such that $\forall\ x \in (a - \delta_\varepsilon, a + \delta_\varepsilon) \cap D$, it holds that

$$|f(x) - f(a)| < \varepsilon.$$

We shall later show that the two definitions are equivalent (Theorem 1.1). First let's see some examples to understand the concept of continuity.

Example 2.1

$f(x) = \sin x$ is continuous for any $x \in \mathbb{R}$.

Thought process: Let a be any real number. The main idea here is to find a suitable constant C_a which may depend on a such that $|\sin x - \sin a| \le C_a|x - a|$. In this problem, we have $C_a = 1$ as $|\sin x - \sin a| \le |x - a|$ (as will see in the formal proof).
Now, if we are using sequential definition, we first take any arbitrary real sequence $\{x_n\}$. The next step is to fix $\varepsilon > 0$, then we choose $N \in \mathbb{N}$ such that for all $n \ge N$, we have $|x_n - a| < \varepsilon/C_a = \varepsilon$ (as $C_a = 1$ here). So, for all $n \ge N$, we must have $|\sin x_n - \sin a| \le |x_n - a| < \varepsilon$.
If we are using ε–δ definition, first we fix $\varepsilon > 0$, then take $\delta \in (0, \varepsilon/C_a] = (0, \varepsilon]$ (as $C_a = 1$ here). So, for all $x \in \mathbb{R}$ such that $|x - a| < \delta$, we must have $|\sin x - \sin a| \le |x - a| < \delta \le \varepsilon$.
Below is how we write this formally.

Solution. First let's try to use the ***sequential definition*** of continuity to prove the above statement.
Let $a \in \mathbb{R}$ be any real number. Take any sequence $\{x_n\} \subset \mathbb{R}$ that converges to a. Fix any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \ge N$, we have $|x_n - a| < \varepsilon$. So, $|f(x_n) - f(a)| = |\sin x_n - \sin a| = 2\left|\cos\left(\frac{x_n+a}{2}\right)\sin\left(\frac{x_n-a}{2}\right)\right| \le |x_n - a| < \varepsilon$. [Here, we

have used $\cos y \leq 1$ and $\sin y \leq y$]. Hence, $f(x_n) \to f(a)$. So, $\sin x$ is continuous for all $x \in \mathbb{R}$. □

Now, let's do the same problem using $\varepsilon - \delta$ ***definition***.
Fix $\varepsilon > 0$. Take $\delta = \varepsilon$ (we can also take $\delta \in (0, \varepsilon]$). Then for all $x \in (a-\delta, a+\delta)$, we have

$$|f(x) - f(a)| = |\sin x - \sin a| = 2\left|\cos\left(\frac{x+a}{2}\right)\sin\left(\frac{x-a}{2}\right)\right| \leq |x-a| < \varepsilon.$$

So, $\sin x$ is continuous for all $x \in \mathbb{R}$. □

Note. So, for all $\varepsilon > 0$, we have found a $\delta_\varepsilon > 0$ (which, in this example, is equal to ε) such that for all $x \in (a - \delta_\varepsilon, a + \delta_\varepsilon)$, it holds $|f(x) - f(a)| < \varepsilon$.

Example 2.2

$f(x) = \sqrt{x}$ is continuous at $x = a$ for every $a \in \mathbb{R}^+$.

Thought process: We will proceed in the same way as in the last problem. Let $a > 0$. We will try to find C_a such that $|f(x) - f(a)| \leq C_a|x-a|$. Note that $|\sqrt{x} - \sqrt{a}| = \left|\frac{x-a}{\sqrt{x}+\sqrt{a}}\right| \leq \frac{1}{\sqrt{a}}|x-a|$. So, here we choose $C_a = \sqrt{a}$. So, while using sequential definition, choose N large enough so that for all $n \geq N$ we have $|x_n - a| < \varepsilon/C_a = \varepsilon\sqrt{a}$. While using $\varepsilon - \delta$ definition, just take $\delta \in (0, \varepsilon/C_a] = (0, \varepsilon\sqrt{a}]$ (we can simply choose $\delta = \varepsilon\sqrt{a}$). Now look at the textbook style proof below.

Solution. (Using *sequential definition*) Let $a \in \mathbb{R}^+$. Take any arbitrary sequence $\{x_n\} \subset \mathbb{R}^+$ that converges to a. Fix $\varepsilon > 0$. $\exists\, N \in \mathbb{N}$ such that $\forall\, n \geq N$, we have $|x_n - a| < \varepsilon\sqrt{a}$. So, $\forall\, n \geq N$, we have

$$|\sqrt{x_n} - \sqrt{a}| = \left|\frac{x_n - a}{\sqrt{x_n} + \sqrt{a}}\right| \leq \frac{1}{\sqrt{a}}|x_n - a| < \varepsilon$$

(The above is true as $\sqrt{x_n} \geq 0$). So, $f(x_n) = \sqrt{x_n} \to \sqrt{a} = f(a)$ as $x_n \to a$. Hence, $f(x) = \sqrt{x}$ is continuous at $x = a$ for every $a \in \mathbb{R}^+$. □

(Using $\varepsilon - \delta$ *definition*) Let $a \in \mathbb{R}^+$. Fix $\varepsilon > 0$. Choose $\delta \in (0, \varepsilon\sqrt{a}]$ (or simply take $\delta = \varepsilon\sqrt{a}$). Then for all $x \in \mathbb{R}^+$ such that $|x - a| < \delta$, we have,

$$|\sqrt{x} - \sqrt{a}| = \left|\frac{x-a}{\sqrt{x}+\sqrt{a}}\right| \leq \frac{1}{\sqrt{a}}|x-a| < \frac{1}{\sqrt{a}} \cdot \delta \leq \varepsilon$$

So, $f(x) = \sqrt{x}$ is continuous at $x = a$ for every $a \in \mathbb{R}^+$. □

Example 2.3

Show that $f(x) = \lfloor x \rfloor \ \forall \ x \in \mathbb{R}$ is continuous at $x = a$ if and only if $a \notin \mathbb{Z}$.

Thought process: Here we need to show that f is continuous at non-integer points and discontinuous at integer points. For the continuity part, what value of δ will work for this problem? Since $a \notin \mathbb{Z}$, we have $\lfloor a \rfloor < a < \lceil a \rceil$. What if we take δ small enough so that $(a - \delta, a + \delta)$ becomes a subset of $(\lfloor a \rfloor, \lceil a \rceil)$? (Think)
Now, how do we prove the discontinuity at a point? One way is to assume that the function is continuous at the point and then find a contradiction. If we are using sequential definition, we can try to construct two sequences which converge to different limits and hence we will arrive contradiction. If we are using $\varepsilon - \delta$ definition, we need to choose a suitable ε such that for any $\delta > 0$, we can get a $x \in (a - \delta, a + \delta)$ such that $|f(x) - f(a)| \geq \varepsilon$. Hence a contradiction!

Solution. (Using *$\varepsilon - \delta$ definition*) Let $a \in \mathbb{R}$. We will divide the solution into the following two cases.

Case 1: $a \notin \mathbb{Z}$.
Fix $\varepsilon > 0$. Choose $\delta \in (0, \min\{a - \lfloor a \rfloor, \lceil a \rceil - a\})$. Then for $x \in (a - \delta, a + \delta) \subset (\lfloor a \rfloor, \lceil a \rceil)$, we have $\lfloor x \rfloor = \lfloor a \rfloor$, therefore, $|f(x) - f(a)| = |\lfloor x \rfloor - \lfloor a \rfloor| = 0 < \varepsilon$.
So, f is continuous at any non-integer point.

Case 2: $a \in \mathbb{Z}$.
Take any $\varepsilon \in (0, 1)$, say $\varepsilon = 3/4$. Then $\forall \ \delta > 0$, take $x = a - \delta$, which is obviously less than a. Then $\lfloor x \rfloor \leq \lfloor a \rfloor - 1$, therefore, $|\lfloor x \rfloor - \lfloor a \rfloor| \geq 1 > 3/4 = \varepsilon$. So, f is not continuous at any integer point. $\square$

(Using *sequential definition*) Let $a \in \mathbb{R}$. Again we will make the same case divisions as above.

Case 1: $a \notin \mathbb{Z}$.
Let $\{x_n\} \subset \mathbb{R}$ be any sequence converging to a. Fix $\varepsilon > 0$. Let $\varepsilon' = \min\{\varepsilon, a - \lfloor a \rfloor, \lceil a \rceil - a\}$. $\exists \ N \in \mathbb{N}$, such that for any $n \geq N$, we have, $x_n \in (a - \varepsilon', a + \varepsilon') \subseteq (\lfloor a \rfloor, \lceil a \rceil)$. Therefore, for $n \geq N$, we have $\lfloor x_n \rfloor = \lfloor a \rfloor$, so, $|f(x_n) - f(a)| = |\lfloor x_n \rfloor - \lfloor a \rfloor| = 0 < \varepsilon$. Hence, $f(x_n) \to f(a)$. So, f is continuous at any non-integer point.

Case 2: $a \in \mathbb{Z}$.
Assume f is continuous at a. Consider the sequence $\{x_n\}$ defined by $x_n = a - \frac{1}{n}$ and $\{y_n\}$ defined by $y_n = a + \frac{1}{n}$. Both the sequences converge to a. $f(x_n) = \lfloor a - \frac{1}{n} \rfloor = a - 1$ for all $n \in \mathbb{N} \implies f(x_n) \to a - 1 \implies f(a) = a - 1$ as f is continuous at a. $f(y_n) = \lfloor a + \frac{1}{n} \rfloor = a$ for all $n \in \mathbb{N} \implies f(y_n) \to a \implies f(a) = a$ as f is continuous at a. Hence, a contradiction! So, f is not continuous at any integer point. $\square$

Theorem 2.1

Let $f : D \to \mathbb{R}$ be a function. Then the following two statements are equivalent:

(1) For every $\varepsilon > 0$, $\exists\, \delta > 0$ such that $|f(x) - f(a)| < \varepsilon$ whenever $x \in (a - \delta, a + \delta) \cap D$.

(2) For any sequence $\{x_n\} \subset D$ that converges to a, it holds that $f(x_n)$ converges to $f(a)$.

Proof. First let's show that (1) $\implies$ (2). Fix $\varepsilon > 0$. $\exists\, \delta > 0$ such that $\forall\, x \in (a-\delta, a+\delta) \cap D$, we have $|f(x) - f(a)| < \varepsilon$. Take a sequence $\{x_n\}$ in D, which converges to a. $\exists\, N \in \mathbb{N}$ such that $\forall\, n \geq N$, we have $x_n \in (a - \delta, a + \delta)$ (as $x_n \to a$). Therefore, $\forall\, n \geq N$, $|f(x_n) - f(a)| < \varepsilon$. Hence, $f(x_n)$ converges to $f(a)$.
Now, we shall show (2) $\implies$ (1). Assume there exists a function f which satisfies property (2) but not property (1), i.e., f has the following properties:

- For any sequence $\{x_n\} \subset D$ that converges to a, it holds that $f(x_n)$ converges to $f(a)$.
- $\exists\, \varepsilon > 0$ such that $\forall\, \delta > 0$, $\exists\, x \in (a - \delta, a + \delta) \cap D$, such that $|f(x) - f(a)| \geq \varepsilon$.

For each $n \in \mathbb{N}$, take $\delta = \frac{1}{n}$. We get a $x_n \in (a - \frac{1}{n}, a + \frac{1}{n})$, such that $|f(x_n) - f(a)| \geq \varepsilon$ $(*)$.
Note that $x_n \to a$, as $a \pm \frac{1}{n} \to a$. So, $\exists N \in \mathbb{N}$, such that $\forall\, n \geq N$, we have $|f(x_n) - f(a)| < \varepsilon$ (by property 2), which contradicts $(*)$. Hence any function which satisfies statement (2), must also satisfy statement (1). $\square$

Theorem 2.2

Between any two reals, there is a rational number.

Proof. Let $a > b$ be two reals. Archimedean property says that there exists $n \in \mathbb{N}$ such that $n(a - b) > 2$. So, $na > \lfloor na \rfloor - 1 > nb \implies a > \frac{\lfloor na \rfloor - 1}{n} > b$. So, $\frac{\lfloor na \rfloor - 1}{n}$ is a rational number between a and b. $\square$

Theorem 2.3

Between any two reals, there is an irrational number.

Proof. Let a, b be two reals. By Theorem 1.2, there is a rational number r between $a + \sqrt{2}$ and $b + \sqrt{2}$. So, $r - \sqrt{2}$ is an irrational number between a and b. $\square$

Theorem 2.4

Given a real number a, there exists a sequence of rationals converging to a.

Proof. By Theorem 1.2, we can find a rational number between a and $a + \frac{1}{n}$, call it r_n. So, by squeeze theorem, $r_n \to a$. Hence, $\{r_n\}$ is a sequence of rationals converging to a. $\square$

Theorem 2.5

Given a real number a, there exists a sequence of irrationals converging to a.

Proof. By Theorem 1.3, we can find a rational number between a and $a + \frac{1}{n}$, call it q_n. So, by squeeze theorem, $q_n \to a$. Hence, $\{q_n\}$ is a sequence of irrationals converging to a. $\square$

Example 2.4 (Dirichlet function)

Show that the following function is discontinuous everywhere

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q} \\ 1 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Thought process: As in Example 1.3, when we use the sequential definition, we need to take two sequences $\{x_n\}$ and $\{y_n\}$ that converge to a, so that their $\{f(x_n)\}$ and $\{f(y_n)\}$ do not converge to the same limit. When we use the $\varepsilon - \delta$ definition, we need to find some $\varepsilon > 0$ such that for any $\delta > 0$, we can find $x \in (a - \delta, a + \delta)$, satisfying $|f(x) - f(a)| \geq \varepsilon$.

Solution. (Using *sequential definition*) Assume f is continuous at $a \in \mathbb{R}$. By Theorem 1.4, there exists a sequence $\{r_n\}$ of rationals that converge to a. Since $f(r_n) = 0 \; \forall \; n \in \mathbb{N}$, we have $f(a) = 0$ $(*)$ (as f is continuous at a). Theorem 1.5 gives us a sequence $\{q_n\}$ of irrationals that converge to a. Since $f(q_n) = 1 \; \forall \; n \in \mathbb{N}$, we have $f(a) = 1$ (since f is continuous at a), contradicting $(*)$. Hence, f is discontinuous at a. $\square$

(Using $\varepsilon - \delta$ *definition*) Assume f is continuous at a.

Case 1: $a \in \mathbb{Q}$.
Take $\varepsilon = 1/2$. $\exists \; \delta > 0$ such that $\forall \; x \in (a - \delta, a + \delta)$, we have $|f(x) - 0| < 1/2$ ($f(a) = 0$ as $a \in \mathbb{Q}$). Take any irrational $x_0 \in (x - \delta, a + \delta)$, then $f(x_0) = 1 > 1/2$, a contradiction! Therefore, f is discontinuous at any rational point.

Case 2: $a \notin \mathbb{Q}$.
Take $\varepsilon = 1/2$. $\exists\, \delta > 0$ such that $\forall\, x \in (a-\delta, a+\delta)$, we have $|f(x)-1| < 1/2$ ($f(a) = 1$ as $a \notin \mathbb{Q}$). Take any rational $x_1 \in (a-\delta, a+\delta)$, then $f(x_1) = 0 \implies |f(x_1)-1| = 1 > 1/2$, again a contradiction! There f is discontinuous at any irrational point. □

Example 2.5

Let

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q} \\ x & \text{if } x \notin \mathbb{Q} \end{cases}$$

Find all $a \in \mathbb{R}$ such that $f(x)$ is continuous at $x = a$.

Thought process: Look at the graph! $y = 0$ and $y = x$ intersect only at the origin. So, our intuition tells us that f is continuous only at 0.

Solution. We claim that f is continuous only at 0. We shall use the sequential definition to prove our claim.

Case 1: $a = 0$.
Take any sequence $\{x_n\}$ that converges to 0. Observe that $0 \leq |f(x_n)| \leq |x_n|$. By Squeeze theorem, $f(x_n) \to 0 = f(0)$. So, f is continuous at 0.

Case 2: $a \neq 0$.
Assume f is continuous at a. Take a sequence $\{x_n\}$ of rationals that converges to a. We have $f(x_n) = 0\, \forall\, n \in \mathbb{N} \implies f(a) = 0$ (since f is continuous at a). Take another sequence $\{y_n\}$ of irrationals that converges to a. We have $f(y_n) = y_n\ \forall\ n \in \mathbb{N} \implies f(a) = a$ (as f is continuous at a). Thus we arrive at a contradiction as $a \neq 0$. Therefore, f is discontinuous at every $a \neq 0$.

Hence, f is continuous only at 0. □

Readers can try solving the above problem using the $\varepsilon - \delta$ definition as an exercise.

Example 2.6

Let

$$f(x) = \begin{cases} 3x & \text{if } x \in \mathbb{Q} \\ x+8 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Find all $a \in \mathbb{R}$ such that $f(x)$ is continuous at $x = a$.

> **Thought process:** Again look at the graphs of $y = 3x$ and $y = x + 8$. They intersect at $(4, 12)$. So, our intuition tells us that the only point of continuity will be at $x = 4$. But how do we prove it? What about taking some suitable functions $g(x)$ and $h(x)$ such that $g(x) \leq f(x) \leq h(x)$ such that $\lim_{x\to 4} g(x) = \lim_{x\to 4} f(x) = \lim_{x\to 4} h(x)$? What "best" functions can we choose?

Solution. We claim that $f(x)$ is continuous only at $x = 4$. We shall present the solution based on the sequential definition.
Note that

$$\min\{3x, x + 8\} \leq f(x) \leq \max\{3x, x + 8\}.$$

We have

$$\min\{3x, x + 8\} = \frac{3x + x + 8 + |3x - x - 8|}{2} = 2x + 4 + |x - 4|$$

and

$$\min\{3x, x + 8\} = \frac{3x + x + 8 - |3x - x - 8|}{2} = 2x + 4 - |x - 4|$$

So,

$$2x + 4 + |x - 4| \leq f(x) \leq 2x + 4 - |x - 4|$$

$\forall\ x \in \mathbb{R}$. Take any sequence $\{x_n\}$ that converges to 4. Therefore,

$$2x_n + 4 - |x_n - 4| \leq f(x_n) \leq 2x_n + 4 + |x_n - 4|.$$

By Squeeze theorem, since $x_n \to 4$, $|x_n - 4| \to 0$, hence, $f(x_n) \to 12 = f(4)$. So, f is continuous at 4.

Now, let $a \neq 4$. We assume that f is continuous at a. Take a rational sequence $\{r_n\}$ and an irrational sequence $\{q_n\}$, both of which converge to a. $f(r_n) = 3r_n\ \forall\ n \in \mathbb{N} \implies f(a) = 3a$. $f(q_n) = q_n + 8\ \forall\ n \in \mathbb{N} \implies f(a) = a + 8$. But since $a \neq 4$, we have $3a \neq a + 8$, a contradiction! Hence, f is discontinuous at $a \neq 4$. □

> **Remark.** In the above solution, we have used a well-known (and easy to prove) result that $\max\{a, b\} = \frac{1}{2}(a + b + |a - b|)$ and $\min\{a, b\} = \frac{1}{2}(a + b - |a - b|)$.

Readers may try solving the above problem using $\varepsilon - \delta$ definition as an exercise.

Example 2.7

Let

$$f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Show that f is continuous at 0.

Solution. Fix $\varepsilon > 0$. Take $\delta = \varepsilon$. $\forall$ x such that $|x| < \delta$, we have,

$$|f(x) - f(0)| = \left|x \sin \frac{1}{x}\right| \leq |x| < \delta = \varepsilon.$$

So, f is continuous at 0. $\square$

Try to prove the same using sequential definition as well.

Example 2.8

Show that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = x^2$ is continuous at every $x \in \mathbb{R}$.

Thought process: We need to find a suitable δ. Suppose, we have a suitable δ that works for this problem. For all $x \in (a - \delta, a + \delta)$, we have $|x^2 - a^2| = |x + a||x - a| \leq (|x| + |a|)|x - a| < (|a| + \delta + |a|)|x - a| = (2|a| + \delta)|x - a|$. We need this to be $< \varepsilon$. We can choose $\delta \leq 1$, then we must have $|x - a| < \varepsilon/(2|a| + 1)$. So, let's take $\delta = \min\{1, \varepsilon/(2|a| + 1)\}$.

Solution. Let's solve this using $\varepsilon - \delta$ definition.
Let $a \in \mathbb{R}$. Fix $\varepsilon > 0$. Take $\delta = \min\left\{1, \frac{\varepsilon}{2|a|+1}\right\}$. $\forall x$ such that $|x - a| < \delta$, we have,

$$\begin{aligned}
|f(x) - f(a)| = |x^2 - a^2| = &|x + a||x - a| \\
&\leq (|x| + |a|)|x - a| \\
&< (|a| + \delta + |a|)|x - a| \\
&= (2|a| + \delta)|x - a| \\
&\leq (2|a| + 1)|x - a| < \varepsilon
\end{aligned}$$

Hence, f is continuous at a. $\square$

Example 2.9 (Lipschitz continuity)

Suppose $f : \mathbb{R} \to \mathbb{R}$ is a function that satisfies

$$|f(x) - f(y)| \leq K|x - y| \quad \forall\ x, y \in \mathbb{R}$$

(Here K is a fixed). Show that $f(x)$ is continuous at $x = a$ $\forall$ $a \in \mathbb{R}$.

Solution. Let $a \in \mathbb{R}$. Take any sequence $\{x_n\}$ that converges to a. Fix $\varepsilon > 0$. $\exists$ $N \in \mathbb{N}$ such that $\forall$ $n \geq N$, we have $|x_n - a| < \varepsilon/K$. Therefore, for $n \geq N$, we have $|f(x_n) - f(a)| \leq K|x_n - a| < \varepsilon$. So, f is continuous at a. $\square$

Now, try using the $\varepsilon - \delta$ definition to prove the above.

Example 2.10

Let $J = \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\}$. Let $a = \frac{1}{k}$ for some $k \in \mathbb{N}$. Let $f : J \to \mathbb{R}$ be any function. Show that f is continuous at a.

> **Thought process:** The thing to note here is that the domain J is a set of isolated points. So, we can make δ small enough so that there is no element of J other than a in the interval $(a - \delta, a + \delta)$.

Solution. Fix $\varepsilon > 0$. Let $J_k := (\frac{1}{k+1}, \frac{1}{k-1})$ if $k \geq 2$ and $(\frac{1}{2}, \frac{3}{2})$ if $k = 1$. Choose $\delta > 0$ such that $(a - \delta, a + \delta) \subset J_k$. So, if $x \in (a - \delta, a + \delta) \cap J$, then $x = a$. So, $|f(x) - f(a)| = 0 < \varepsilon$ whenever $x \in (a - \delta, a + \delta) \cap J$. Hence, f is continuous at a. $\square$

Note. We can also show that δ should be in the interval $(0, \frac{1}{k} - \frac{1}{k+1})$. (why?)

> **Remark.** The graph of the above function CANNOT be drawn without lifting the pen, yet it is a continuous function. It is a misconception that a function whose graph cannot be drawn without lifting the pen is discontinuous.

Example 2.11

$f : J \to \mathbb{R}$ be continuous at c with $f(c) \neq 0$. Show that $\exists\ \delta > 0$ such that $|f(x)| > |f(c)|/2$ for all $x \in (c - \delta, c + \delta) \cap J$.

Solution. Fix $\varepsilon = |f(c)|/2$. Since f is continuous at c, we can find $\delta > 0$ such that $\forall\ x \in (c - \delta, c + \delta) \cap J$, we have

$$\begin{aligned} |f(x) - f(c)| < |f(c)|/2 &\implies |f(c)| - |f(x)| < |f(x) - f(c)| < |f(c)|/2 \\ &\implies |f(x)| > |f(c)|/2. \end{aligned}$$

$\square$

§2.1 Exercise 1

Problem 2.1 (Algebraic properties of continuous functions)**.** Suppose that $f(x)$ and $g(x)$ are continuous at $x = a$. Then, the following functions are also continuous at $x = a$: $f(x) + g(x)$, $f(x) - g(x)$, $cf(x)$ (where c is a constant), $1/f(x)$ (provided $f(a) \neq 0$), $f(x)g(x)$, $|f(x)|$, $\max\{f(x), g(x)\}$, $\min\{f(x), g(x)\}$.

> **Remark.** The result of the above problem is fundamental and will appear in nearly every problem in the following sections. Keep this result in mind!

Problem 2.2. Show that $f(x) = x^n \ \forall \ x \in \mathbb{R}$ is continuous ($n \in \mathbb{N}$ is fixed). Further show that any polynomial function is continuous on $\mathbb{R}$.

Problem 2.3. Suppose that $f : A \to B$ and $g : B \to C$ (so that $(g \circ f)(x) = g(f(x))$ is well-defined). If $f(x)$ is continuous at $x = a$ and $g(y)$ is continuous at $y = f(a)$, show that $h(x) = g(f(x))$ must be continuous at $x = a$.

***Problem 2.4** (Thomae function)**.** Let $f : (0, 1) \to \mathbb{R}$ be given by

$$f(x) = \begin{cases} 0 & \text{if } x \notin \mathbb{Q} \\ 1/q & \text{if } x = p/q, \quad p, q \in \mathbb{N}, \ \gcd(p, q) = 1 \end{cases}$$

Show that f is continuous at all irrational points and discontinuous at any rational point in $(0, 1)$.

***Problem 2.5.** Study the continuity of the following function:

$$f(x) = \begin{cases} |x| & \text{if } x \notin \mathbb{Q} \text{ or } x = 0, \\ qx/(q + 1) & \text{if } x = p/q, \quad p, q \in \mathbb{N}, \ \gcd(p, q) = 1 \end{cases}$$

***Problem 2.6.** Determine all a_n and b_n for which the function defined by

$$f(x) = \begin{cases} a_n + \sin \pi x & \text{if } x \in [2n, 2n + 1], \ n \in \mathbb{Z}, \\ a_n + \cos \pi x & \text{if } x \in (2n - 1, 2n), \ n \in \mathbb{Z} \end{cases}$$

is continuous on $\mathbb{R}$.

Problem 2.7. Study the continuity of the function $f(x) = \lfloor x^2 \rfloor \sin \pi x$ for $x \in \mathbb{R}$.

Problem 2.8. Consider $J = \left\{ \frac{1}{n} \mid n \in \mathbb{N} \right\} \cup \{0\}$. Let $f : J \to \mathbb{R}$ be a function. Define $y_n := f(\frac{1}{n})$ and $y := f(0)$. Show that f is continuous if and only if $y_n \to y$.

Problem 2.9 (Banach Contraction Mapping Theorem)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a function, and assume that there is a constant $k \in (0, 1)$ such that

$$|f(x) - f(y)| \leq k|x - y| \ \ \forall \ x, y \in \mathbb{R}.$$

(a) Show that f is continuous on $\mathbb{R}$.

(b) Pick some point $x_0 \in \mathbb{R}$ and construct the sequence $x_{n+1} = f(x_n)$. Show that the resulting sequence $\{x_n\}$ is a Cauchy sequence and converges to some point $x^* \in \mathbb{R}$.

(c) Show that x^* is a fixed point of f (i.e., $f(x^*) = x^*$).

(d) Show that f has a unique fixed point.

§3 Intermediate Value Theorem

Suppose you have a continuous function $f(x)$, and there exists a, b such that $f(a)$ and $f(b)$ are of different signs. Also, assume the entire interval $[a, b]$ is a subset of the domain of f. Can you draw the graph of f without cutting the x-axis in $[a, b]$? (Try to convince yourself by drawing multiple graphs that you cannot do so!). This is what the Intermediate Value Theorem has to say.

Note. It is advisable to first practice using the theorem by going through the solved examples below before reading its proof.

Theorem 3.1 (Intermediate Value Theorem)
Suppose $f : [a, b] \to \mathbb{R}$ be a continuous function such that $f(a) < f(b)$. Then for any $k \in (f(a), f(b))$, there exists $c_k \in (a, b)$ such that $f(c_k) = k$.

Proof. There are many proofs of this theorem. The proof presented here uses Nested Intervals Theorem and the "midpoint algorithm".
Before we proceed, if you don't know the Nested Intervals Theorem, here is the statement (you should try to prove this on your own):

> For each n, let $I_n = [a_n, b_n]$ be a (non-empty) bounded interval of real numbers such that
> $$I_1 \supset \ I_2 \supset I_3 \supset \cdots \supset I_n \supset I_{n+1} \supset \cdots$$
> and $\lim\limits_{n\to 0}(b_n - a_n) = 0$. Then $\bigcap\limits_{n=1}^{\infty} I_n$ contains only one point.

Now let's begin with the proof...

Let $f : [a, b] \to \mathbb{R}$ be continuous such that $f(a) < f(b)$. Let $k \in (f(a), f(b))$. Assume to the contrary that there is no point c in (a, b) for which $f(c) = k$. Define $I_0 := [a, b]$ and $x_1 = \frac{1}{2}(a + b)$. If $f(x_1) > k$, define $a_1 = a$ and $b_1 = x_1$, else, define $a_1 = x_1$ and $b_1 = b$. So, we have $k \in (f(a_1), f(b_1))$ and we perform the same process again. Define $x_2 = \frac{1}{2}(a_1 + b_1)$. If $f(x_2) > k$, define $a_2 = a_1$ and $b_2 = x_2$, else, define $a_2 = x_2$ and $b_2 = b_1$. We repeat this infinitely. At each step, we define $x_{n+1} = \frac{1}{2}(a_n + b_n)$. If $f(x_{x+1}) > k$, we define $a_{n+1} = a_n$ and $b_{n+1} = x_{n+1}$, else we define $a_{n+1} = x_{n+1}$ and $b_{n+1} = b_n$. Also, define $I_n = [a_n, b_n]$.

Note that $I_{n+1} \subset I_n$ and $b_{n+1} - a_{n+1} = \frac{1}{2}(b_n - a_n)$ for each $n \geq 1$ (Do you see why?). So, the lengths of the intervals converge to 0. So, by Nested Intervals Theorem, the intersection of all the intervals consists of exactly one real number, call it c'. Therefore, $a_n \to c'$ and $b_n \to c'$. Also, we have $f(a_n) < k < f(b_n)$. Since f is continuous, we have

$f(a_n) \to f(c')$ and $f(b_n) \to f(c')$ as $n \to \infty$. Therefore by Squeeze Theorem, we have $f(c') = k$, which is a contradiction! Hence there must exist a point $c \in [a,b]$ such that $f(c) = k$. □

Corollary 3.1

Suppose $f : [a,b] \to \mathbb{R}$ be continuous such that $f(a) < 0 < f(b)$. Then there must exist $c \in (a,b)$ such that $f(c) = 0$.

The above corollary often becomes useful while solving problems.

Now let's see some examples...

Example 3.1

Prove that there exists $x \in \mathbb{R}$ such that $xe^x = 1$.

Solution. Define $h(x) := xe^x - 1$. It suffices to show that the equation $h(x) = 0$ has a solution. Note that $h(0) = -1$ and $h(1) = e - 1$. So, we have $h(0) < 0 < h(1)$. Since $h(x)$ is continuous, the Intermediate Value Theorem (essentially Corollary 2.1) guarantees that there exists $x_0 \in (0,1)$ such that $h(x_0) = 0$. □

Remark. To show that $h(x)$ is continuous, you can first try to show that e^x is continuous. Then use the result from Problem 1.1 of Exercise 1 to show that xe^x is also continuous (product of two continuous functions is continuous) and hence conclude that $xe^x - 1$ is also continuous.

Remark. We could also have taken $h(x) = xe^x$. In that case, we would have $0 = h(0) < 1 < h(1) = e$. Then, by the Intermediate Value Theorem, there must exist $x_0 \in (0,1)$ such that $h(x_0) = 1$, i.e., $x_0 e^{x_0} = 1$.

Example 3.2

Assume that $f, g : [a,b] \to \mathbb{R}$ are continuous and $f(a) < g(a)$ and $f(b) > g(b)$. Prove that there exists $c \in (a,b)$ such that $f(c) = g(c)$.

Solution. Define $h(x) := f(x) - g(x)\ \forall\ x \in [a,b]$. Clearly $h(x)$ is a continuous function. Observe that $h(a) = f(a) - g(a) < 0$ and $h(b) = f(b) - g(b) > 0$. Hence, by Intermediate Value Theorem, $\exists\ c \in (a,b)$ such that $h(c) = 0$, i.e., $f(c) = g(c)$. □

Theorem 3.2 (Brouwer's Fixed Point Theorem (1-dimensional case))

Let $f : [a, b] \to [a, b]$ be continuous. Show that $\exists\ c \in [a, b]$ such that $f(c) = c$, i.e., f has a fixed point in $[a, b]$.

Proof. Define $h(x) := f(x) - x$ for all $x \in [a, b]$. It suffices to show that there exists $c \in [a, b]$ such that $h(c) = 0$. We have $h(a) \geq 0$ and $h(b) \leq 0$ (as $a \leq f(x) \leq b$ for all $x \in [a, b]$). If $h(a) = 0$ or $h(b) = 0$, then we are already done. So, assume $h(a) > 0$ and $h(b) < 0$. Since h is continuous, by Intermediate Value Theorem, $\exists\ c \in (a, b)$ such that $h(c) = 0$, i.e., $f(c) = c$. □

Example 3.3

Show that the function $f : \mathbb{R} \to \mathbb{R}$, defined by $f(x) = (x - a)^2(x - b)^2 + x$ has the value $\frac{a+b}{2}$ in its range.

Solution. If $a = b$, then we are already done (why?). So, WLOG assume $a < b$. Note that $f(a) = a$ and $f(b) = b$. So, we have $f(a) < \frac{a+b}{2} < f(b)$. Also observe that f is a continuous function (see Problem 1.2 of Exercise 1). So, applying Intermediate Value Theorem, $\exists\ c \in (a, b)$ such that $f(c) = \frac{a+b}{2}$. □

Example 3.4

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and periodic with period $T > 0$. Prove that there eixsts x_0 such that $f\left(x_0 + \frac{T}{2}\right) = f(x_0)$.

Solution. Define $h(x) := f\left(x + \frac{T}{2}\right) - f(x)$ for all x. Notice that $h(0) = f\left(\frac{T}{2}\right) - f(0)$ and $h\left(\frac{T}{2}\right) = f(T) - f\left(\frac{T}{2}\right) = f(0) - f\left(\frac{T}{2}\right)$ (since f has a period T, we have $f(T) = f(0)$). If $f\left(\frac{T}{2}\right) = f(0)$, then we are already done! So assume $f\left(\frac{T}{2}\right) \neq f(0)$. Therefore, $h(0)$ and $h\left(\frac{T}{2}\right)$ have different signs. Since h is continuous, by Intermediate Value Theorem, there exists $x_0 \in \left(0, \frac{T}{2}\right)$, such that $f\left(x_0 + \frac{T}{2}\right) = f(x_0)$. □

Example 3.5

Suppose $f \in C([0, 2])$ be a function with $f(0) = f(2)$. Show that $\exists\ x_1, x_2 \in [0, 2]$ such that $x_1 - x_2 = 1$ and $f(x_1) = f(x_2)$.

Solution. Assume $f(0) \neq f(1)$ (otherwise we are done). Let $h(x) := f(x + 1) - f(x)$ for all $x \in [0, 1]$. It suffices to show that there exists $c \in [0, 1]$ for which $h(c) = 0$. Clearly h is continuous and we also have $h(0) = f(1) - f(0)$ and $h(1) = f(2) - f(1) = f(0) - f(1)$.

Since $f(0) \neq f(1)$, we can say $h(0)$ and $h(1)$ are of different signs. Hence, by the Intermediate Value Theorem, $\exists\ c \in (0,1)$ such that $f(c+1) = f(c)$. $\square$

Note. In the above problem $C([0,2])$ denotes the set of all functions continuous on $[0,2]$. Some authors prefer to use the notation $C^0([0,2])$ instead.

The next theorem is a very useful and pretty well-known result related to polynomials. There are various ways of proving the result. Here, we shall present a proof that uses Intermediate Value Theorem.

Theorem 3.3

Let $P(x)$ be a polynomial with real coefficients and suppose that the degree of $P(x)$ is odd. Show that the equation $P(x) = 0$ must have at least one real root.

Proof. WLOG, assume P is monic. Let $P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$. For $x \neq 0$, we can write $P(x) = x^n\left(1 + \frac{a_{n-1}}{x} + \frac{a_{n-2}}{x^2} + \cdots + \frac{a_0}{x^n}\right)$. Let $m \in \mathbb{Z} - \{0\}$, then $\left|\frac{a_k}{m^k}\right| \leq \left|\frac{a_k}{m}\right|$ for any $1 \leq k \leq n$. By triangle inequality, we have,

$$\begin{aligned}\left|\frac{a_{n-1}}{m} + \frac{a_{n-2}}{m^2} + \cdots + \frac{a_0}{m^n}\right| &\leq \left|\frac{a_{n-1}}{m}\right| + \left|\frac{a_{n-2}}{m^2}\right| + \cdots + \left|\frac{a_0}{m^n}\right| \\ &\leq \left|\frac{a_{n-1}}{m}\right| + \left|\frac{a_{n-2}}{m}\right| + \cdots + \left|\frac{a_0}{m}\right| = \frac{C}{|m|},\end{aligned}$$

where $C = |a_{n-1}| + |a_{n-2}| + \cdots + |a_0|$.
Choose $|m|$ large enough so that $\frac{C}{|m|} < \frac{1}{2}$. So, So, we have

$$\left|\frac{a_{n-1}}{m} + \frac{a_{n-2}}{m^2} + \cdots + \frac{a_0}{m^n}\right| < \frac{1}{2}$$

$$\implies \frac{1}{2} < 1 + \frac{a_{n-1}}{m} + \frac{a_{n-2}}{m^2} + \cdots + \frac{a_0}{m^n} < \frac{3}{2}.$$

So, $x^n/2 < P(x) < 3x^n/2$ for $x \in \{m, -m\}$. WLOG assume m to be positive. This means $P(m) > m^n/2 > 0$, and $P(-m) < 3(-m)^n/2 = -3m^n/2 < 0$. So, Intermediate Value Theorem tells that there must be $x_0 \in (-m, m)$ such that $P(x_0) = 0$. $\square$

Note. The above proof is taken from the book *A Basic Course in Real Analysis* by S. Kumerasan and Ajit Kumar. Here is another proof of the above theorem.

Proof. Let $\{x_k\}$ be a sequence of positive reals such that $x_k \to \infty$ as $k \to \infty$. Also consider another sequence $\{y_k\}$ of negative reals such that $y_n \to -\infty$ as $k \to \infty$. Let

$P(x) = x^n + a_{n-1}x^{n-1} + \cdots + a_0$ (WLOG, we assume P is monic). From what we have already learned in the chapter sequence, we know,

$$\lim_{k\to\infty} \frac{x_k^d}{x_k^n} = \begin{cases} 1 & \text{if } d = n \\ 0 & \text{if } 0 \le d < n \end{cases}$$

So,

$$\lim_{k\to\infty} \frac{P(x_k)}{x_k^n} = 1.$$

Therefore, $\exists\ N \in \mathbb{N}$ such that $|P(x_N)/x_N^n - 1| < 1/2 \implies x_N^n/2 < P(x_N)$. So, $P(x_N) > 0$.

In the same way, we can say $\exists\ M \in \mathbb{N}$ such that $|P(y_M)/y_M^n - 1| < 1/2 \implies P(y_M) < 3y_M^n/2 < 0$.

Hence, by Intermediate Value Theorem, $\exists\ x_0 \in (y_M, x_N)$ such that $P(x_0) = 0$ (since P is continuous). ☐

Remark. In the above proof, in place of $1/2$, we could have taken any real in $(0, 1)$.

Example 3.6

For $a_0 < b_0 < a_1 < b_1 < \ldots < a_n < b_n$, show that all the roots of the polynomial

$$P(x) = \prod_{k=0}^{n}(x + a_k) + 2\prod_{k=0}^{n}(x + b_k) \ \ \forall\ x \in \mathbb{R}$$

are real.

Solution. Firstly note that P has at most $n+1$ real roots (why?). Now, observe that $(b_k - a_p)$ is negative for each $n \ge p > k$, and is positive for $0 \le p \le k$. So, $\prod_{k=0}^{n}(b_k - a_p)$ is negative if p is odd, and is positive if p is even. Similarly, $\prod_{k=0}^{n}(a_k - b_p)$ is negative if p is even, and is positive if p is odd.

Now, note that

$$P(-a_p) = 2\prod_{k=0}^{n}(b_k - a_p)$$

and

$$P(-b_p) = \prod_{k=0}^{n}(a_k - b_p)$$

for each $p \in \{0, 1, 2, \ldots, n\}$. So, we have, $P(-a_p) < 0 < P(-b_p)$ for p odd, and $P(-b_p) < 0 < P(-a_p)$ for p even. Thus, by Intermediate Value Theorem, there exists

$\alpha_i \in (-b_i, -a_i)$ such that $P(\alpha_i) = 0$ for each $i = 0, 1, 2, \ldots, n$. So, all the $n+1$ roots of P are real. □

§3.1 Exercise 2

Problem 3.1. Prove that the equation $(1-x)\cos x = \sin x$ has atleast one solution in $(0, 1)$.

Problem 3.2. Let $f : [0,1] \to [0,1]$ be a continuous function such that $f(0) = 0$ and $f(1) = 1$. Prove that $\exists\ c \in (0,1)$ such that $c^2 + (f(c))^2 = 1$.

Problem 3.3. Show that $x^4 + 5x^3 - 7 = 0$ has atleast two real roots.

Problem 3.4. Suppose that $f : [a,b] \to \mathbb{R}$ is a continuous function that takes rational values only. What can you say about f?

Problem 3.5. Prove that the equation $x^n + x^{n-1} + \cdots + x - 1 = 0$ has a unique positive solution. Let a_n denote this solution. Show that $a_n \leq 1$. Show that $a_n^{n+1} - 2a_n + 1 = 0$. Also, find the limit of a_n as $n \to \infty$.

Problem 3.6. Let $f \in C([0,2])$. Prove that $\exists\ x_1, x_2 \in [0,2]$ such that $x_1 - x_2 = 1$ and

$$f(x_1) - f(x_2) = \frac{1}{2}(f(2) - f(0)).$$

Problem 3.7. Let $f : (a,b) \to \mathbb{R}$ be continuous. Prove that given $x_1, x_2, \ldots, x_n \in (a,b)$, there exists $x_0 \in (a,b)$ such that

$$f(x_0) = \frac{1}{n}(f(x_1) + f(x_2) + \cdots + f(x_n)).$$

Problem 3.8. For each $n \in \mathbb{N}$, let $f \in C([0,n])$ be such that $f(0) = f(n)$. Prove that there exists $x_1, x_2 \in [0,n]$ satisfying $x_2 - x_1 = 1$ and $f(x_2) = f(x_1)$.

Problem 3.9. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous and periodic with period $T > 0$. Prove that there exists x_0 such that $f(x_0 + \pi) = f(x_0)$.

Problem 3.10. Let $P(x) = a_0 + a_1 x + \cdots + a_n x^n$ be a polynomial, where n is even. If $a_0 a_1 < 0$, show that $P(x) = 0$ has atleast two real roots.

***Problem 3.11.** Prove that the trigonometric polynomial

$$f(x) = a_0 + a_1 \cos x + a_2 \cos 2x + \cdots + a_n \cos nx,$$

where the coefficients are all real and $|a_0| + |a_1| + \cdots + |a_{n-1}| \leq a_n$, has at least $2n$ zeros in the interval $[0, 2\pi)$.

***Problem 3.12.** Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous and strictly decreasing function. Prove that the system $x = f(y),\ y = f(z),\ z = f(x)$ has a unique solution.

***Problem 3.13.** Suppose f and g are real continuous functions defined on $\mathbb{R}$, that satisfy $f(g(x)) = g(f(x))$ for all $x \in \mathbb{R}$. Prove that if the equation $f^2(x) = g^2(x)$ has a solution, then the equation $f(x) = g(x)$ also has a solution.

§4 Intermediate Value Property

Definition 4.1

A function $f : [a, b] \to \mathbb{R}$ with $f(a) < f(b)$ is said to have *intermediate value property* if for every $k \in (f(a), f(b))$, there exists $c_k \in (a, b)$ such that $f(c_k) = k$.

We have already learned in the last section that any continuous function on a closed interval has this property. But is there any discontinuous function which possesses intermediate value property?

The answer turns out to be YES! Consider the following function:

$$f(x) = \begin{cases} \sin \frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Can you see that the function is discontinuous only at 0? So, on any closed interval not containing 0, the function is continuous, so it has intermediate value property. Now consider an interval $[a, b]$ such that $0 \in [a, b]$. We will show that $f([a, b]) = [-1, 1]$.

Let $k \in [-1, 1]$. $\exists\ x_0 \in [-\pi/2, \pi/2]$, such that $\sin x_0 = k$. So, $\forall\ n \in \mathbb{Z}$, we have $\sin(x_0 + 2\pi n) = k$. Atleast one of a, b is not equal to zero, say $b > 0$ (if $a < 0$, take $x_n = \frac{1}{x_0 - 2\pi n}$ in the lines that follow). Consider the sequence $\{x_n\}_{n\geq 1}$ defined by $x_n = \frac{1}{x_0 + 2\pi n}$. Clearly, the sequence $\{x_n\}_{n\geq 1}$ converges to 0. So, $\exists\ N \in \mathbb{N}$ such $0 < x_N < b$. Therefore, $f(x_N) = \sin(x_0 + 2\pi N) = \sin(x_0) = k$. Hence, for any $k \in [-1, 1]$ we can find a $x \in [a, b]$ such that $f(x) = k$. Therefore, f has intermediate value property.

Below is the graph of the above function. See how the function oscillates around the point 0.

(**Optional reading**) Okay, so now we know a function that is discontinuous at exactly one point yet satisfies the intermediate value property. But we can go further! Let's construct a function that has this property and is discontinuous at infinitely many points.

Before we do that, let's first define something called the *Cantor set*.

Consider the closed interval $[0, 1]$, and label it C_0. In the next step, divide the interval $[0, 1]$ into three parts: $\left[0, \frac{1}{3}\right]$, $\left(\frac{1}{3}, \frac{2}{3}\right)$, and $\left[\frac{2}{3}, 1\right]$, and delete the middle open interval

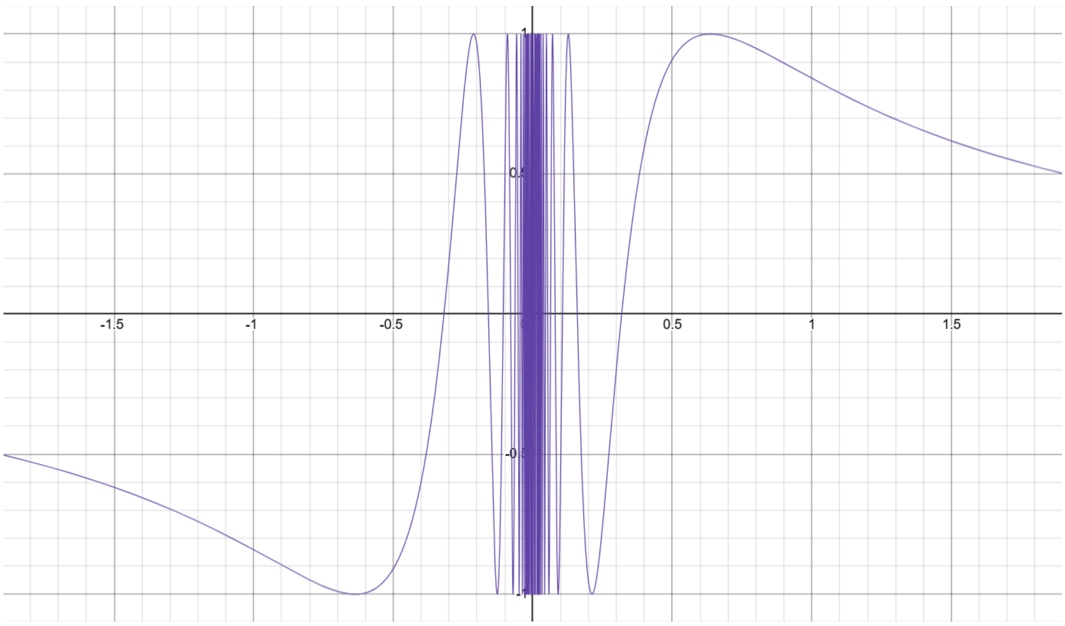

Figure 4.1: Graph of $f(x)$ as defined above, showing the oscillations around 0.

$\left(\frac{1}{3}, \frac{2}{3}\right)$. Let C_1 denote the union of the two remaining intervals, $\left[0, \frac{1}{3}\right]$ and $\left[\frac{2}{3}, 1\right]$. In the following step, take the two intervals in C_1, divide each into three equal parts of length $\frac{1}{9}$, and delete the middle open intervals $\left(\frac{1}{9}, \frac{2}{9}\right)$ and $\left(\frac{7}{9}, \frac{8}{9}\right)$. Let C_2 be the union of the remaining intervals.

Continuing this process, at the nth step we remove the union of the open middle thirds from each of the remaining 2^{n-1} intervals. We denote by C_n the union of the 2^n closed intervals that remain, each of length 3^{-n}.

Finally, we define the *Cantor set* as:

$$C = \bigcap_{n=0}^{\infty} C_n.$$

> **Remark.** A point $x \in [0, 1]$ belongs to C if and only if its base-3 (ternary) expansion contains only the digits 0 and 2. (see here for a detailed explanation)

Note that if $\{(a_i, b_i)\}_{i=1}^{\infty}$ denote the sequence of the open intervals removed during the construction of C, then

$$C = [0, 1] \setminus \bigcup_{n=1}^{\infty} (a_i, b_i).$$

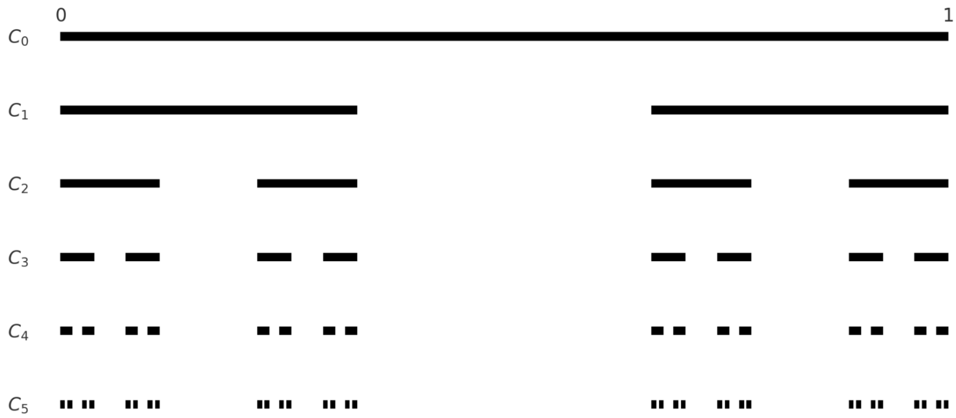

Figure 4.2: Construction of the Cantor Set

Define the function $g : [0, 1] \to \mathbb{R}$ by

$$g(x) = \begin{cases} 0 & \text{if } x \in C \\ \frac{2(x-a_i)}{b_i-a_i} - 1 & \text{if } x \in (a_i, b_i), \quad i = 1, 2, \dots \ . \end{cases}$$

Firstly, note that $g(x)$ is linear on each removed interval (a_i, b_i), mapping (a_i, b_i) onto $(-1, 1)$. So, g is continuous on every removed interval and we also have $g((a_i, b_i)) = g([0, 1]) = (-1, 1)$ for each $i = 1, 2, \dots$.

We claim that g is discontinuous at every point in C and it enjoys the intermediate value property.
Consider any interval $[a, b] \subset [0, 1]$. If $[a, b] \cap C = \emptyset$, then $[a, b]$ must be a subinterval of some removed interval (a_i, b_i). Since g is continuous on (a_i, b_i), we have that g enjoys intermediate value property on $[a, b]$. Now, if there is some $x \in [a, b] \cap C$, then there exists $n \in \mathbb{N}$ and $k \in \{0, 1, \dots, 3^n - 1\}$, such that $x \in \left[\frac{k}{3^n}, \frac{k+1}{3^n}\right] \subset [a, b]$. Then the open middle third of $\left[\frac{k}{3^n}, \frac{k+1}{3^n}\right]$ must be one of the removed intervals, say (a_m, b_m). So, $g((a_m, b_m)) = (-1, 1)$, hence g enjoys intermediate value property on $[a, b]$ (since, $g([a, b]) = (-1, 1) = g((a_m, b_m)))$.

Thus g is discontinuous at each C, but enjoys intermediate value property.

§4.1 Exercise 3

Problem 4.1. Show that the following function is discontinuous but satisfies the intermediate value property:

$$f(x) = \begin{cases} 2x \sin \frac{1}{x^2} - \frac{2}{x} \cos \frac{1}{x^2} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Problem 4.2. Suppose that f and g have the intermediate value property on some closed bounded interval I. Is it necessary that $f+g$ also has the intermediate value property on I?

***Problem 4.3** (RNMO 2004). Let $f:(a,b)\to\mathbb{R}$ be a function such that for any $x\in(a,b)$ there exists a non-degenerate interval $[a_x,b_x]$ with $a\le a_x\le x\le b_x\le b$ such that f is constant in $[a_x,b_x]$. Find all such function f satisfying intermediate value property.

***Problem 4.4.** Prove that if $f:\mathbb{R}\to\mathbb{R}$ has intermediate value property and $f^{-1}(\{q\})$ is closed for every rational q, then f is continuous.

§5 Extreme Value Theorem

Let's first solve a problem that will help motivate the proofs of the upcoming theorems.

Example 5.1

Does there exist a function $f:[0,1]\to[0,1)$ which is surjective and continuous?

Solution. The answer is NO. Suppose such a function exists. Define $y_n := 1-\frac{1}{n}\in[0,1)$ for all $n\ge 1$. Since f is surjective, $\exists\ x_n\in[0,1]$ such that $f(x_n)=y_n$. $\{x_n\}$ is a bounded sequence, so by Bolzano-Weierstrass Theorem, it has a convergent subsequence, call it $\{x_{n_k}\}$, which converges to some $c\in[0,1]$. Since, f is continuous, $f(x_{n_k})\to f(c)$ as $x_{n_k}\to c$. But, $f(x_{n_k})=y_{n_k}=1-\frac{1}{n_k}\to 1$. So, $f(c)=1$ for some $c\in[0,1]$, which is a contradiction! □

Theorem 5.1 (Weierstrass Theorem)

Suppose that $f:[a,b]\to\mathbb{R}$ is a continuous function. Then f must be bounded.

Proof. Suppose that f is unbounded. $\exists\ x_n\in[a,b]$ such that $|f(x_n)|>n$ for each $n\in\mathbb{N}$. Since $\{x_n\}$ is bounded, we can find a convergent subsequence $\{x_{n_k}\}$ which converges to some $\alpha\in[a,b]$ (by Bolzano-Weierstrass Theorem). Then, due to continuity of $|f|$, $|f(x_{n_k})|\to|f(\alpha)|$. But $|f(x_{n_k})|>n_k$, So, $\{|f(x_{n_k})|\}$ diverges, which is a contradiction. Hence, f is bounded. □

Note. First-time readers may choose to skip the proof of the next theorem, but should make sure to read it later.

Theorem 5.2 (Extreme Value Theorem)

Suppose that $f : [a, b] \to \mathbb{R}$ is a continuous function. Then there exists $c, d \in [a, b]$ such that $f(c) \geq f(x) \geq f(d)$ holds for every $x \in [a, b]$.

Proof. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. Theorem 4.1 guarantees that f has a supremum and an infimum. We will show that the supremum is the maximum of f, the proof of the minimum will be similar. Let $s = \sup\limits_{x \in [a,b]} f(x)$. Suppose f does not attain its supremum anywhere, i.e., $\nexists\ x \in [a, b]$ such that $f(x) = s$. Then for each $n \in \mathbb{N}$, there exists x_n such that $s - \frac{1}{n} < f(x_n) \leq s$. Consider a convergent subsequence $\{x_{n_k}\}$ of $\{x_n\}$ which converges to some $c \in [a, b]$ (the existence of such a subsequence is guaranteed by Bolzano-Weierstrass Theorem). So, by Squeeze Theorem and due to continuity of f, $\lim\limits_{k \to \infty} f(x_{n_k}) = s \implies f\left(\lim\limits_{k \to \infty} x_{n_k}\right) = s \implies f(c) = s$, which is a contradiction. Thus, f attains a maximum. □

The result below is a significant consequence of the Extreme Value Theorem and the Intermediate Value Theorem. You are encouraged to attempt the proof on your own before reading further.

Theorem 5.3

A continuous function maps a closed bounded interval into a closed bounded interval. In other words, if $f : [a, b] \to \mathbb{R}$ is continuous, then image of f is $[c, d]$ for some $c < d$.

Proof. By the Extreme Value Theorem, f attains its maximum and its minimum, i.e., $\exists\ c, d \in \mathbb{R}$ such that $c \leq f(x) \leq d$ for all $x \in [a, b]$. Let $f(m_*) = c$ and $f(m^*) = d$. Now, by Intermediate Value Theorem, for each $k \in (c, d)$, there exists x_k between m_* and m^* such that $f(x_k) = k$. This completes the proof. □

§5.1 Exercise 4

Problem 5.1. Try to find another proof of Theorem 4.2 using Nested Intervals Theorem and "midpoint algorithm".

Problem 5.2. Let $f : [a, b] \to \mathbb{R}$ be a continuous function with the property that for every $x \in [a, b]$, there exists $y \in [a, b]$ such that $|f(y)| \leq \frac{1}{2}|f(x)|$. Show that there exists $c \in [a, b]$ such that $f(c) = 0$.

Problem 5.3. Let $f, g : [0, 1] \to [0, 1]$ be continuous such that

$$\sup_{x \in [0,1]} f(x) = \sup_{x \in [0,1]} g(x)$$

Prove that $\exists\, t \in [0,1]$ such that $f(t)^2 + 3f(t) = g(t)^2 + 3g(t)$

***Problem 5.4** (IMC 2011). Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function. A point x is called a shadow point if there exists a point $y \in \mathbb{R}$ with $y > x$ such that $f(y) > f(x)$. Let $a < b$ be real numbers and suppose that

- all points of the open interval $I = (a,b)$ are shadow points;
- a and b are not shadow points.

Prove that

(a) $f(x) \leq f(b)$ for all $a < x < b$;

(b) $f(a) = f(b)$.

§6 Monotonicity and fixed points

You probably know everything about monotone functions, but still, for the sake of completeness, here are the definitions:

Definition 6.1

We call a function f *monotone*, if f is either weakly increasing or weakly decreasing, i.e., either $f(x) \geq f(y)$ whenever $x > y$ or we have $f(x) \leq f(y)$ whenever $x > y$.

We call a function f *strictly monotone*, if f is either strictly increasing or strictly decreasing, i.e., either $f(x) > f(y)$ whenever $x > y$, or, $f(x) < f(y)$ whenever $x > y$.

Definition 6.2

Let $f : D \to \mathbb{R}$ be a function. We say that $x_0 \in D$ is a *fixed point* of f if we have $f(x_0) = x_0$.

Here is a very useful result related to monotone functions. It could have been included in the section on the Intermediate Value Property, but it seemed more appropriate to present it in a separate section dedicated to monotone functions. You can refer to the diagram of Figure 5.1 to get a better understanding of the theorem.

Theorem 6.1

If $f : [a,b] \to \mathbb{R}$ is injective and has intermediate value property, then f is strictly monotone.

Proof. For the sake of contradiction, suppose that there exists an injective function f which has intermediate value property but is not strictly monotone. So, there must exist $a < b < c$ such that either $f(b) < f(a), f(c)$ or $f(b) > f(a), f(c)$. WLOG, assume the first inequality holds (otherwise work with $-f$). Now, if $f(a) < f(c)$, then, we have $f(b) < f(a) < f(c)$. Since f has intermediate value property, $\exists\ d \in (b, c)$ such that $f(d) = f(a) \in (f(b), f(c))$, contradicting the injectivity of f. So, we must have $f(a) > f(c)$, but then $f(b) < f(c) < f(a)$. Since f has intermediate value property, $\exists\ d' \in (a, b)$ such that $f(d') = f(c) \in (f(b), f(a))$, again contradicting the injectivity of f. This completes the proof. $\square$

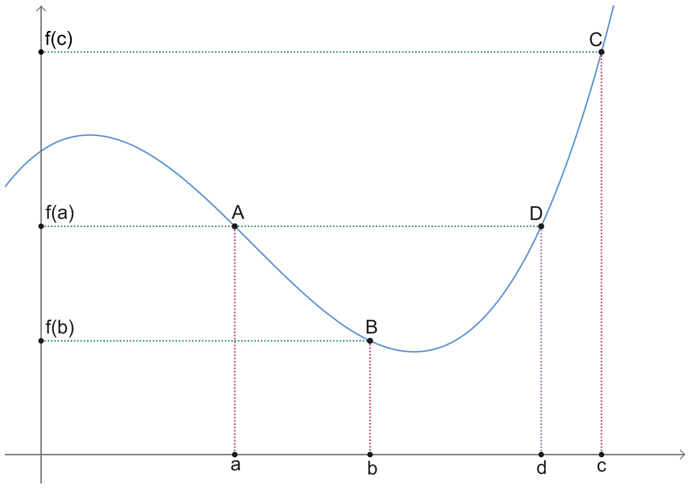

Figure 6.1: A graph showing a continuous function which is not strictly monotone cannot be injective.

Corollary 6.1

If $f : [a, b] \to \mathbb{R}$ is injective and continuous, then it is strictly monotone.

Theorem 6.2

Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function that maps open intervals to open intervals. Prove that f is monotone.

Proof. Suppose f is not monotone. If f is constant in an open interval, then we get a contradiction! WLOG, assume there exist points $a < b < c \in \mathbb{R}$ such that $f(a), f(c) < f(b)$ (otherwise, work with $-f$). Consider the interval $[a, c]$. Clearly, f cannot attain a maximum at its end points a and c (do you see why?). So, it must attain a maximum at a point $d \in (a, c)$ (by Extreme Value Theorem). But then, $f((a, c))$ cannot be an open interval, as open intervals cannot have a maximum (but can have a supremum!). $\square$

Theorem 6.3

Suppose $f : \mathbb{R} \to \mathbb{R}$ be a continuous function satisfying $f^n(x) = x$ for all $x \in \mathbb{R}$ (for some $n \in \mathbb{N}$), then

(a) $f(x) = x$, $x \in \mathbb{R}$, if f is strictly increasing,

(b) $f^2(x) = x$, $x \in \mathbb{R}$, if f is strictly decreasing.

Proof. Firstly, note that if f is not injective, then there exists $a \neq b$ such that $f(a) = f(b)$. But then $f^n(a) = f^{n-1}(f(a)) = f^{n-1}(f(b)) = f^n(b) \implies a = b$, which is a contradiction (This is a very common trick, so remember this!). Therefore, f is injective. So, by Corollary 5.1, f is either strictly increasing or strictly decreasing.

Case 1: Suppose f is strictly increasing. Suppose there exists $\alpha \in \mathbb{R}$ such that $f(\alpha) \neq \alpha$. WLOG, assume $f(\alpha) > \alpha$. Since f is strictly increasing, we have $f(x) > f(y)$ whenever $x > y$. Hence we have, $f^2(\alpha) > f(\alpha)$ as $f(\alpha) > \alpha$. Then we must have $f^3(\alpha) > f(\alpha) > \alpha$ and continuing in this way, we have (by induction) $f^n(\alpha) > f^{n-1}(\alpha) > \alpha$, which is a contradiction.

Case 2: Suppose f is strictly decreasing. Suppose there exists $\beta \in \mathbb{R}$ such that $f^2(\beta) \neq \beta$. WLOG, assume $f^2(\beta) > \beta$. Since f is strictly decreasing, we must have $f(x) < f(y)$ whenever $x > y$. Hence, $f^2(x) > f^2(y)$ whenever $x > y$. Therefore, $f^{2n}(\beta) > \beta$. But, $f^n(\beta) = \beta$ leads to $f^{2n}(\beta) = f^n(\beta) = \beta$. Hence, a contradiction! □

Example 6.1 (Picard's Sequence)

Let $f : [a, b] \to [a, b]$ be a continuous function. Let $x_0 \in [a, b]$. Consider the sequence $\{x_n\}$ defined by $x_n = f(x_{n-1})$ for each positive integer n. Further assume that $\{x_n\}$ converges to some $x^* \in [a, b]$. Prove that x^* is a fixed point of f, i.e, $f(x^*) = x^*$.

Proof. Since f is continuous, we have $\lim\limits_{n\to\infty} f(x_n) = f\left(\lim\limits_{n\to\infty} x_n\right)$. But we have $f(x_n) = x_{n+1}$, thus $\lim\limits_{n\to\infty} x_{n+1} = f\left(\lim\limits_{n\to\infty} x_n\right) \implies x^* = f(x^*)$. □

Example 6.2

Suppose $f : [a, b] \to [a, b]$ be a function such that $|f(x) - f(y)| \leq \frac{1}{2}|x - y|$ for all $x, y \in [a, b]$. Show that there exists $\xi \in [a, b]$ such that $f(\xi) = \xi$.

Solution. Firstly, note that f is continuous on $[a, b]$ (see Example 1.9). Fix any $x_0 \in [a, b]$. Define $a_n = f^n(x_0)$ for each $n \geq 1$. and $a_0 = x_0$. Note that

$$|f^n(x_0) - f^{n-1}(x_0)| \leq \frac{1}{2}|f^{n-1}(x_0) - f^{n-2}(x_0)| \leq \cdots \leq \frac{1}{2^{n-1}}|f(x_0) - x_0|.$$

So, we have

$$|a_n - a_{n-1}| \leq \frac{1}{2^{n-1}}C,$$

where $C = |a_1 - a_0|$. Clearly, $\{a_n\}_{n\geq 0}$ is a contractive sequence, and hence is also Cauchy sequence. So, $\{a_n\}_{n\geq 0}$ converges to some ξ (here we use the fact that every Cauchy sequence is convergent). Since, $a \leq a_n \leq b$ for all $n \geq 0$, we have $\xi \in [a, b]$. Now, since, $a_{n+1} = f(a_n)$ for all $n \geq 0$, taking the limit as $n \to \infty$ and using the fact that f is continuous, we get

$$\xi = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} f(a_n) = f\left(\lim_{n\to\infty} a_n\right) = f(\xi).$$

□

§6.1 Exercise 5

Problem 6.1. Let $f : \mathbb{R} \to \mathbb{R}$ be a surjective continuous function that takes any value at most twice. Prove that f is strictly monotone.

Problem 6.2. Prove that there is no continuous function $f : \mathbb{R} \to \mathbb{R}$ that takes every value exactly twice.

Problem 6.3. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous such that $|f(x) - f(y)| \geq |x - y|\ \forall\ x, y \in \mathbb{R}$. Prove that f is surjective.

Problem 6.4. Assume $f : \mathbb{R} \to \mathbb{R}$ satisfy the condition $f(f(x)) = -x$ for all $x \in \mathbb{R}$. Show that f cannot be continuous.

***Problem 6.5.** Find all functions $f : \mathbb{R} \to \mathbb{R}$which have the intermediate value property and such there is $n \in \mathbb{N}$ for which $f^n(x) = -x$ for all $x \in \mathbb{R}$, where f^n denotes the nth iteration of f.

***Problem 6.6.** Let $f : [a, b] \to [a, b]$ be an arbitrary function.

(a) Prove that if f is continuous, then f has a fixed point. (same as Theorem 2.2)

(b) Show that the result remains true if f is non-decreasing (i.e., monotonically increasing).

(c) Find a decreasing function f with no fixed points.

***Problem 6.7.** Let $f : [0, 1] \to [0, 1]$ be a function such that $|f(x) - f(y)| \le |x - y|$ for all $x, y \in [0, 1]$. Prove that the set of all fixed points of f is either a single point or an interval.

***Problem 6.8.** Let $f : [a, b] \to [a, b]$ be a continuous function. Let $x_0 \in [a, b]$. Consider the sequence $\{x_n\}$ defined by $x_n = f(x_{n-1})$ for each positive integer n. Prove that $\{x_n\}$ converges if and only if $(x_{n+1} - x_n)$ converges to 0.

***Problem 6.9.** Let $f : [a, b] \to [a, b]$ be a function satisfying $|f(x)-f(y)| \le |x-y| \; \forall \, x, y \in [a, b]$. Define the sequence $\{x_n\}_{n\ge1}$ by $x_1 \in [a, b]$ and for all $n \ge 1$, $x_{n+1} = (x_n + f(x_n))/2$. Prove that $\{x_n\}$ converges to to some fixed point of f.

§7 Limits of functions

§7.1 Limit points

Let's begin with the terminologies first...

Definition 7.1 (Limit point or Accumulation point or Cluster point of a set)
A point a is a said to be a *limit point* of $D \subseteq \mathbb{R}$, if every deleted $\varepsilon-$neighborhood of a (i.e., $(a - \varepsilon, a + \varepsilon) \setminus \{a\}$) contains some element of D.

We can also define a limit point of a set using a sequential definition.

Definition 7.2 (Limit point or Accumulation point or Cluster point of a set)
A point a is said to be a *limit point* of $D \subseteq \mathbb{R}$ if there exists a sequence $\{x_n\} \subseteq D$ such that $x_n \to a$ and $x_n \ne a$ for all $n \ge 1$.

First let's show that the above two definitions are equivalent. (You should first try proving this yourself before looking into the proof)

Theorem 7.1
Let $a \in \mathbb{R}$ and $D \subseteq \mathbb{R}$. Then the following statements are equivalent:

(1) Every deleted $\varepsilon-$neighborhood of a contains some element of D.

(2) There exists a sequence $\{x_n\} \subseteq D$ such that $x_n \to a$ and $x_n \ne a$ for all $n \ge 1$.

Proof. First let's show that (2) $\implies$ (1). Since $x_n \to a$ and $x_n \ne a \; \forall \; n \ge 1$, we can say that for any $\varepsilon > 0$, there exists $N \in \mathbb{N}$ such that for all $n \ge N$, we have $x_n \in (a - \varepsilon, a + \varepsilon) \setminus \{a\}$. So, for every $\varepsilon > 0$, the deleted $\varepsilon-$neighborhood of a contains some elements of D, i.e., $D \cap (a-\varepsilon, a+\varepsilon) \setminus \{a\} \neq \emptyset$.

Now let's show that (1) $\implies$ (2). Since every deleted $\varepsilon-$neighborhood of a contains some element of D, we can say that there exists $x_n \in D \cap \left(a - \frac{1}{n}, a + \frac{1}{n}\right) \setminus \{a\}$ for each $n \in \mathbb{N}$. Clearly, the sequence x_n converges to a and $x_n \neq a$ for all $n \in \mathbb{N}$. This completes the proof. ☐

Now let's try to understand the meaning of limit point of a set through some examples.

Example 7.1

Consider $D = (0, 1)$.

(a) Is $\frac{1}{2}$ a limit point of D?

(b) Is 0 a limit point of D?

(c) Is $\frac{3}{2}$ a limit point of D?

Solution. (a) Clearly, for any $\varepsilon > 0$, $(0,1) \cap \left(\frac{1}{2} - \varepsilon, \frac{1}{2} + \varepsilon\right) \setminus \{\frac{1}{2}\} \neq \emptyset$. so, every deleted $\varepsilon-$neighborhood of $\frac{1}{2}$ contains some element of D. So, by Definition 6.1, $\frac{1}{2}$ is a limit point of D.
Alternatively, we can also show this using the sequential definition. Consider the sequence $\{x_n\}$ defined by $x_n = \frac{1}{2} + \frac{1}{n+2}$ for all $n \geq 1$. Clearly, $x_n \in (0,1) \setminus \{\frac{1}{2}\}$ for all $n \geq 1$ and $x_n \to \frac{1}{2}$. Hence, we arrive at the same conclusion.
In the same way, we can also show that any $r \in (0, 1)$ is a limit point of D.

(b) Consider the sequence $\{x_n\}$ defined by $x_n = \frac{1}{n}$. Clearly, $x_n \in (0,1) \setminus \{\frac{1}{2}\}$ and $x_n \to \frac{1}{2}$. Thus 0 is a limit point of D. Try to show this using the other definition. Similarly, we can also show that 1 is also a limit point of D.

Note. Note that 0 or 1 are not in $D = (0, 1)$, but still they are limit points of D. So, it is not necessary that the limit points has to be an element of D.

(c) Note that $(0,1) \cap \left(\frac{3}{2} - \frac{1}{2}, \frac{3}{2} + \frac{1}{2}\right) \setminus \{\frac{3}{2}\} = \emptyset$. So, by Definition 6.1, the deleted $\varepsilon-$neighborhood with $\varepsilon = \frac{1}{2}$ does not contain any element of D. So, $\frac{3}{2}$ is not a limit point of D. Now try to arrive at the same conclusion using the other definition too. In the same way, we can also show that any $r \notin [0, 1]$ is not a limit point of D.

Example 7.2

Consider $D = \left\{\frac{1}{n} \mid n \in \mathbb{N}\right\}$. Find all limit points of D.

Solution. We claim that the only limit point of $D = \left\{\frac{1}{n} \mid n \in \mathbb{N}\right\}$ is 0.

Clearly, 0 is a limit point as it is the limit of the sequence $x_n = \frac{1}{n}$, all whose terms are non-zero and are in D. But why is no other point a limit point of the set D? Lets consider an arbitrary positive real r. We choose the smallest $a, b \in \mathbb{N}$ such that $\frac{1}{a} < r < \frac{1}{b}$. Now, take $\varepsilon = \min\left\{r - \frac{1}{a}, \frac{1}{b} - r\right\}$. Clearly, in the deleted $\varepsilon-$neighborhood of r (with ε as set before), there is no element of D. So, r cannot be a limit point of D. Now, any negative real r', take $\varepsilon' = |r'|$. Clearly, in the deleted $\varepsilon'-$neighborhood of r', there is no element of D. Hence, any negative real cannot be a limit point of D. Therefore, the only limit point of D is 0.

Example 7.3

Show that the set of all limit points of the set $\mathbb{Q}$ of rationals is $\mathbb{R}$.

Solution. Clearly, by Theorem 1.4, for any real a, there is a sequence of rationals (none of whose terms is equal to a), which converges to a. Hence, the set of all limit points of $\mathbb{Q}$ is $\mathbb{R}$. □

Now, we are ready to formally define what is meant by the "limit of a function".

§7.2 Definitions of limit

Definition 7.3 ($\varepsilon - \delta$ definition of limit)

Suppose $f : D \to \mathbb{R}$ ($D \subseteq \mathbb{R}$) be a function. Let a be a limit point of D. If there exists $L \in \mathbb{R}$ such that $\forall\, \varepsilon > 0$, $\exists\, \delta > 0$ such that $x \in D \cap (a - \delta, a + \delta) \setminus \{a\} \implies |f(x) - L| < \varepsilon$, then we say that the *limit of f at a exists and is equal to L* and we write

$$\lim_{x \to a;\, x \in D} f(x) = L.$$

Definition 7.4 (Sequential definition of limit)

Suppose $f : D \to \mathbb{R}$ ($D \subseteq \mathbb{R}$) be a function. Let a be a limit point of D. If there exists $L \in \mathbb{R}$ such that for every sequence $\{x_n\} \subseteq D$, such that $x_n \to a$ and $x_n \neq a\ \forall\, n \geq 1$, we have $f(x_n) \to L$, then we say that the *limit of f at a exists and is equal to L* and we write

$$\lim_{x \to a;\, x \in D} f(x) = L.$$

We say that f *converges* to L at a if the limit of f at a exists and equals L. If f does not converge to any real number at a, we say that f *diverges* at a and leave $\lim\limits_{x \to a;\, x \in D} f(x)$ undefined.

> **Remark.** We only consider the limit of a function at a limit point of its domain. For any other point, it is not worth it to define the concept of a limit (do you see why?)

Note. Whenever the notation $\lim\limits_{x \to a} f(x)$ appears, it is understood that x approaches a through values in the domain of f. Moreover, unless otherwise specified, the domain of a function is taken to be its *largest real domain* (often called the "natural domain") — that is, the set of all real numbers for which the formula defining $f(x)$ yields a real value.

Now, let's see some examples.

Example 7.4

Let $f(x) = x \sin \frac{1}{x} \ \forall \ x \in \mathbb{R} \setminus \{0\}$. Find $\lim\limits_{x \to 0} \ f(x)$.

Solution. Let's first do this by using the $\varepsilon - \delta$ definition. The domain of $f(x) = x \sin \frac{1}{x}$ is $\mathbb{R} \setminus \{0\}$. Try to convince yourself that 0 is a limit point of $\mathbb{R} \setminus \{0\}$. Now, for any $\varepsilon > 0$, choose $\delta = \varepsilon$. Then, for $x \in (-\delta, \delta) \setminus \{0\}$, we have $|f(x)| = \left|x \sin \frac{1}{x}\right| \leq |x| < \delta = \varepsilon$. Hence, $\lim\limits_{x \to 0} f(x) = 0$.

Now we shall do the same using the sequential definition. Take any sequence $\{x_n\} \in \mathbb{R} \setminus \{0\}$ that converges to 0 (there must exist such a sequence as 0 is a limit point of the domain of f). Then for any $\varepsilon > 0$, $\exists \ N \in \mathbb{N}$ such that $\forall \ n \geq N$, we have $|x_n| < \varepsilon$. So, $|f(x_n)| = \left|x_n \sin \frac{1}{x_n}\right| \leq |x_n| < \varepsilon$ for each $n \geq N$. Hence, $f(x_n) \to 0 \implies \lim\limits_{x \to 0} f(x) = 0$.

Example 7.5

Let $D = \left\{\frac{1}{n} \mid n \in \mathbb{N}\right\}$. Let $f : D \to \mathbb{R}$ be a function defined by $f(x) = x \sin \frac{1}{x} \ \forall \ x \in D$. Find $\lim\limits_{x \to 0;\, x \in D} f(x)$.

Solution. Note that 0 is a limit point of D (see Example 6.2). Now, proceeding as in the previous example, we get the same result, i.e., $\lim\limits_{x \to 0;\, x \in D} f(x) = 0$

Example 7.6

Let $f : \mathbb{R} \to \mathbb{R}$ be a function defined by

$$f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0. \end{cases}$$

Find $\lim\limits_{x \to 0} \ f(x)$.

Solution. Using $\varepsilon - \delta$ definition, for any $\varepsilon > 0$, take $\delta = \varepsilon$, then for $x \in (-\delta, \delta) \setminus \{0\}$, we have $|f(x)| < \varepsilon$ (as in Example 6.4). So, $\lim_{x \to 0} f(x) = 0$.

The sequential definition yields the same result: for any sequence $(x_n) \subset \mathbb{R} \setminus \{0\}$, we again estimate:

$$|f(x_n)| \leq |x_n| \to 0,$$

confirming $f(x_n) \to 0 \implies \lim_{x \to 0} f(x) = 0$.

Note. Notice that although $f(0)$ is defined as 1, the limit of $f(x)$ as $x \to 0$ is 0, not 1. This illustrates a key concept: the limit of a function at a point depends on the values of $f(x)$ "near" that point, not necessarily on the value of f "at" the point itself.

Before we explore more examples, let's first learn a few theorems in this section.

Theorem 7.2

The limit of a function at a point, if exists, must be unique.

Proof. Let a be a limit point of D. Suppose $f : D \to \mathbb{R}$ has two limits at a, say l_1, l_2. Fix $\varepsilon = |l_1 - l_2|/2$, which is strictly greater than 0. Using the $\varepsilon - \delta$ definition, we must have the following:

- $\exists\ \delta_1 > 0$ such that $0 < |x - a| < \delta_1,\ x \in D \implies |f(x) - l_1| < \varepsilon/2$.
- $\exists\ \delta_2 > 0$ such that $0 < |x - a| < \delta_2,\ x \in D \implies |f(x) - l_2| < \varepsilon/2$.

Let $\delta = \min\{\delta_1, \delta_2\}$. So, for all $x \in D$ such that $0 < |x - a| < \delta$, we have $|f(x) - l_1| < \varepsilon/2$ and $|f(x) - l_2| < \varepsilon/2 \implies |l_1 - l_2| \leq |f(x) - l_1| + |f(x) - l_2| < \varepsilon/2 + \varepsilon/2 = \varepsilon = |l_1 - l_2|/2$, which is a contradiction. This completes the proof. $\square$

Theorem 7.3

If $\lim_{x \to a} f(x) = L$, then $\exists\ \delta > 0$ such that f is bounded within $(a - \delta, a + \delta)$.

Proof. Let $f : D \to \mathbb{R}$ be such that a is a limit point of D and f converges to L at a. Fix $\varepsilon > 0$. There exists $\delta > 0$ such that for $x \in D \cap (a - \delta, a + \delta) \setminus \{a\}$, we have $f(x) \in (L - \varepsilon, L + \varepsilon)$, which means f is bounded. This completes the proof. $\square$

Theorem 7.4 (Algebra of finite limits)
If $\lim_{x\to a;\, x\in D} f(x) = L_1$ and $\lim_{x\to a;\, x\in D} g(x) = L_2$, then

(1) $\lim_{x\to a;\, x\in D} (f(x) \pm g(x)) = L_1 \pm L_2$

(2) $\lim_{x\to a;\, x\in D} f(x)g(x) = L_1 L_2$

(3) $\lim_{x\to a;\, x\in D} \alpha f(x) = \alpha L_1$, where $\alpha \in \mathbb{R}$

(4) Assuming $L_1 \neq 0$, then $\lim_{x\to a;\, x\in D} \dfrac{1}{f(x)} = \dfrac{1}{L_1}$

Proof. See Problem 6.2. □

Theorem 7.5 (Squeeze Theorem or Sanswich Theorem)
Let $f : D \to \mathbb{R}$ be function and let a be a limit point of D. Suppose $\exists\, \delta > 0$ such that

$$f(x) \leq g(x) \leq h(x)$$

for all $x \in (a-\delta, a+\delta)\setminus\{a\}$. Further, assume that $\lim_{x\to a;\, x\in D} f(x) = L = \lim_{x\to a;\, x\in D} h(x)$. Then $\lim_{x\to a;\, x\in D} g(x)$ exists and equals L.

Proof. Fix $\varepsilon > 0$. There exists $\delta_1 > 0$ such that for $x \in D \cap (a-\delta_1, a+\delta_1) \setminus \{a\}$, we have $|f(x) - L| < \varepsilon$. There exists $\delta_2 > 0$ such that for $x \in D \cap (a-\delta_2, a+\delta_2) \setminus \{a\}$, we have $|h(x) - L| < \varepsilon$. Take $\delta = \min\{\delta_1, \delta_2\}$. Then for $x \in D \cap (a-\delta, a+\delta) \setminus \{a\}$, we have

$$-\varepsilon < f(x) - L \leq g(x) - L \leq h(x) - L < \varepsilon$$

So, $|f(x) - L| < \varepsilon$. Hence, $\lim_{x\to a;\, x\in D} g(x)$ exists and equals L. □

Example 7.7

Find $\lim_{x\to a;\, x\in\mathbb{R}\setminus\{0\}} x \sin \dfrac{1}{x}$ using Squeeze Theorem.

Solution. Observe that $-|x| < x \sin \frac{1}{x} < |x|$ for all $x \in \mathbb{R} \setminus \{0\}$ and $\lim_{x\to 0;\, x\in\mathbb{R}\setminus\{0\}} (\pm|x|) = 0$. So, by Squeeze Theorem, we can say that $\lim_{x\to a;\, x\in\mathbb{R}\setminus\{0\}} x \sin \dfrac{1}{x}$ exists and equals 0.

Example 7.8

Find $\lim_{x \to 0} \frac{x}{a} \left\lfloor \frac{b}{x} \right\rfloor, \quad a, b > 0.$

Solution. Note that $\frac{b}{x} - 1 < \left\lfloor \frac{b}{x} \right\rfloor \leq \frac{b}{x}$. So, $\frac{b}{a} - \frac{x}{a} < \frac{x}{a} \left\lfloor \frac{b}{x} \right\rfloor \leq \frac{b}{a}$. Also, we have $\lim_{x \to 0} \left(\frac{b}{a} - \frac{x}{a} \right) = \frac{b}{a}$. So, by Squeeze Theorem, we get $\lim_{x \to 0} \frac{x}{a} \left\lfloor \frac{b}{x} \right\rfloor = \frac{b}{a}$.

Theorem 7.6

If $\lim_{x \to a;\, x \in D} f(x) = \alpha$. Let g be a function defined on an open interval containing α and is continuous at α. Let E be the set of points in D for which $g \circ f$ is defined. Then $\lim_{x \to a;\, x \in E} (g \circ f)(x)$ exists and equals $g(\alpha)$.

Proof. Choose $\delta > 0$ such that $(\alpha - \varepsilon, \alpha + \varepsilon) \subseteq$ domain (g) (clearly, there exists such a ε, as g is defined on an open interval containing α). Let $\{x_n\} \subseteq D \setminus \{a\}$ be any sequence that converges to a. As $f(x_n) \to \alpha$, there exists $N \in \mathbb{N}$ such that for all $n \geq N$, we have $f(x_n) \in (\alpha - \varepsilon, \alpha + \varepsilon)$. So, for each $n \geq N$, the function $(g \circ f)(x_n)$ is defined. Now, since g is continuous, we have,

$$\lim_{n \to \infty;\, n \geq N} (g \circ f)(x_n) = \lim_{n \to \infty;\, n \geq N} g(f(x_n)) = g\left(\lim_{n \to \infty;\, n \geq N} f(x_n) \right) = g(\alpha).$$

□

Example 7.9

Find $\lim_{x \to 0} \frac{\sin x}{x}$.

Solution. Here, we assume that the domain of the function $\frac{\sin x}{x}$ is its natural domain, i.e., $\mathbb{R} \setminus \{0\}$. We know $\sin x \leq x \leq \tan x$ for all $x \neq 0$. So, $\cos x \leq \frac{\sin x}{x} \leq 1$. Now, $\lim_{x \to 0} \cos x = 1$. Hence, by Squeeze Theorem, we have $\lim_{x \to 0} \frac{\sin x}{x} = 1$.

§7.3 Infinite limits

Note that according to Definition 6.3 and Definition 6.4, the limit of a function at a point, if exists, must be finite. But, the statement $\lim_{x \to 0} \frac{1}{x^2} = +\infty$ certainly makes intuitive sense. We can define the infinite limits as follows:

Definition 7.5 (Infinite limit)

Let $f : D \to \mathbb{R}$ be a function and let a be a limit point of D. If $\forall\ M > 0$, $\exists \delta > 0$ such that $\forall\ x \in D \cap (a - \delta, a + \delta) \setminus \{a\}$ we have $f(x) > M$, then we say that the limit of f at a is $+\infty$ and we write $\lim\limits_{x\to a} f(x) = +\infty$.

Similarly we define $\lim\limits_{x\to a} f(x) = -\infty$.

§7.4 One-sided limits

Consider the function $f(x) = \lfloor x \rfloor$. Clearly the limit of the function as x approaches 0 does not exist (you can verify this by taking two sequences, one in $(-1, 0)$ and the other in $(0, 1)$, both approaching 0; you will find the functional limit of the sequences are different). But, if we let x approach 0 from the left side of 0, i.e., $(-\infty, 0)$, then we can see that $f(x)$ approaches -1. Similarly, if we let x approach 0 from the right side of 0, i.e., $(0, \infty)$, then $f(x)$ approaches 0. So, the notion of one-sided limit becomes important here. Here is how we define the one-sided limits:

Definition 7.6 (Left-hand limit)

Let $f : D \to \mathbb{R}$ be a function and let a be a limit point of D. If there exists $L \in \mathbb{R}$ such that $\forall\ \varepsilon > 0$, $\exists\ \delta > 0$ such that $x \in D \cap (a, a + \delta) \implies |f(x) - L| < \varepsilon$, then we say that the *right-hand limit of f at a exists and is equal to L* and we write:

$$\lim_{x\to a^+} f(x) = L \text{ or } \lim_{x\searrow a} f(x).$$

Definition 7.7 (Right-hand limit)

Let $f : D \to \mathbb{R}$ be a function and let a be a limit point of D. If there exists $L \in \mathbb{R}$ such that $\forall\ \varepsilon > 0$, $\exists\ \delta > 0$ such that $x \in D \cap (a - \delta, a) \implies |f(x) - L| < \varepsilon$, then we say that the *left-hand limit of f at a exists and is equal to L* and we write:

$$\lim_{x\to a^+} f(x) = L \text{ or } \lim_{x\nearrow a} f(x).$$

Try to express the above definitions in the sequential form.

Theorem 7.7

Let $f : D \to \mathbb{R}$ be a function and let a be a limit point of D. Then $\lim\limits_{x\to a} f(x)$ exists and equals L if and only if $\lim\limits_{x\to a^+} f(x) = \lim\limits_{x\to a^-} f(x) = L$.

Proof. Suppose $\lim\limits_{x\to a} f(x) = L$, then $\forall\ \varepsilon > 0$, $\exists\ \delta > 0$ such that $\forall\ x \in D \cap (a - \delta, a + \delta) \setminus \{a\}$, we have $|f(x) - L| < \varepsilon$. So, $\forall\ x \in D \cap (a - \delta, a)$ and $\forall\ x \in D \cap (a, a + \delta)$ we have

$|f(x) - L| < \varepsilon$, which implies $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L$.

Now, let's prove the other direction. We have $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = L$. Fix $\varepsilon > 0$. There exists $\delta_1, \delta_2 > 0$ such that:

$$x \in D \cap (a - \delta_1, a) \implies |f(x) - L| < \varepsilon,$$
$$x \in D \cap (a, a + \delta_2) \implies |f(x) - L| < \varepsilon.$$

Take $\delta = \min\{\delta_1, \delta_2\}$. So, for all $x \in D \cap (a - \delta, a + \delta) \setminus \{a\}$ we have $|f(x) - L| < \varepsilon$. Hence, $\lim_{x \to a} f(x) = L$. □

> **Remark.** If $a \in D$ is not a limit point of D, then there must be some neighborhood of a, which contains no other point of D. Such a point is called an isolated point of the set D. Although the limit of a function cannot be defined at such a point, the function still remains continuous at any isolated point of its domain.

The following theorem highlights how continuity relies on the behavior of a function's limits.

Theorem 7.8

Let $f : D \to \mathbb{R}$ be a function. Let $a \in D$ be a limit point of D. Then $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = f(a)$ if and only if f is continuous at a.

Proof. If $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = f(a)$, then, for each $\varepsilon > 0$ there exist $\delta_1 > 0$ and $\delta_2 > 0$ such that:

$$x \in D \cap (a - \delta_1, a] \implies |f(x) - f(a)| < \varepsilon,$$
$$x \in D \cap [a, a + \delta_2) \implies |f(x) - f(a)| < \varepsilon.$$

Let $\delta = \min\{\delta_1, \delta_2\}$. Then whenever $x \in D \cap (a - \delta, a + \delta)$, we have $|f(x) - f(a)| < \varepsilon$. This establishes $\lim_{x \to a} f(x) = f(a)$, i.e. f is continuous at a.

Now, we shall prove the other direction. If f is continuous at a, then for each $\varepsilon > 0$, there exists $\delta > 0$ such that for all $x \in D \cap (a - \delta, a + \delta)$, we have $|f(x) - f(a)| < \varepsilon$. Hence, $\lim_{x \to a} f(x) = f(a)$. So, by Theorem 6.7, we conclude that $\lim_{x \to a^+} f(x) = \lim_{x \to a^-} f(x) = f(a)$. □

Definition 7.8

We say that a function f is *discontinuous* at a point a (where a is a limit point of its domain D and $a \in D$) if any of the following occur:

(1) $\lim_{x \to a} f(x)$ does not exist;

(2) $\lim_{x \to a} f(x)$ exists but is not equal to $f(a)$.

In particular, for the function $f(x) = \lfloor x \rfloor$, this definition correctly identifies a discontinuity at 0, since

$$\lim_{x \to 0^+} f(x) = 0 \text{ and } \lim_{x \to 0^-} f(x) = -1.$$

Thus the overall $\lim_{x \to 0} f(x)$ does not exist, so f is discontinuous at 0.

§7.5 Limits at infinity

Until now, we have discussed what it means for a function $f : D \to \mathbb{R}$ to have a limit at x_0 where x_0 is a real number. Now, consider the function $f(x) = \frac{1}{x}$. Notice (from the graph maybe) that as we let x approach ∞, the value of $f(x)$ approaches 0. So, it makes sense to define the limit at $\pm\infty$.

First, we need to need a notion of what it means for $+\infty$ and $-\infty$ to be limit points.

Definition 7.9 (Infinite limit points)

Let $D \subseteq \mathbb{R}$. We say that $+\infty$ is a limit point of D iff for every $M > 0$, there exists $x \in D$, which is greater than $x > M$; we say that $-\infty$ is a limit point of D is for every $M < 0$, there exists $x \in D$ such that $x < M$.

In other words, $+\infty$ is a limit point of D is $\sup(D) = +\infty$ and $-\infty$ is a limit point of D is $\inf(D) = -\infty$.

Now we shall try to give a meaning to "*limit of a function at* $+\infty$".

Definition 7.10 (Limit at $\pm\infty$)

Let D be a subset of $\mathbb{R}$ with $+\infty$ as a limit point. Let $f : D \to \mathbb{R}$ be a function. If there exists L such that given any $\varepsilon > 0$, there exists $M > 0$ such that for all $x \in D \cap (M, +\infty)$, we have $|f(x) - L| < \varepsilon$, then we say

$$\lim_{x \to \infty} f(x) = L.$$

Similarly, we can impart meaning to "limit at $-\infty$".

> **Remark.** You may see that all the definitions can be unified if we work on the extended real line, i.e., $\mathbb{R} \cup \{-\infty, +\infty\}$, instead of $\mathbb{R}$. But here we are going to deal with functions defined on $\mathbb{R}$.

§7.6 Discontinuity

We classify discontinuity of a function at a point into the following three categories:

(1) **Removable discontinuity:** When $\lim_{x\to a^+} f(x) = \lim_{x\to a^-} f(x) \neq f(a)$.
Example: Consider the function

$$f(x) = \begin{cases} x \sin \frac{1}{x} & \text{if } x \neq 0 \\ 1 & \text{if } x = 0. \end{cases}$$

Both the left hand and right hand limits at 0 are 0, but $f(0) = 1$. So, f has removable discontinuity at 0.

(2) **Jump discontinuity:** When both $\lim_{x\to a^+} f(x)$ and $\lim_{x\to a^-} f(x)$ exist, but $\lim_{x\to a^+} f(x) \neq \lim_{x\to a^-} f(x)$.
Example: Consider the function

$$f(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1 & \text{if } x \geq 0. \end{cases}$$

At 0, the left hand limit is 0 and the right hand limit is 1. So, the function has jump discontinuity at 0.

(3) **Essential discontinuity:** When at least one of $\lim_{x\to a^+} f(x)$ and $\lim_{x\to a^-} f(x)$ does not exist.
Example: Consider the function

$$f(x) = \begin{cases} \frac{1}{x} & \text{if } x > 0 \\ 0 & \text{if } x \leq 0. \end{cases}$$

At 0, the left hand limit is 0, but the right hand limit does not exist. So, the function has essential discontinuity at 0.

Theorem 7.9

One-sided limits must exist for monotone functions.

Proof. Let $f : D \to \mathbb{R}$ be a monotone function. We will show that $\lim\limits_{x \to a^-} f(x)$ exists for every limit point a of D. WLOG, assume f is increasing. So, the set $S_- = \{f(x) \mid x < a\}$ is bounded above and hence has a supremum, call it L. Since, L is the supremum of S_-, for every $\varepsilon > 0$, $L - \varepsilon$ is not an upper bound of S_-. So, for any $\varepsilon > 0$, there exists $\delta > 0$ such that $f(a - \delta) > L - \varepsilon$. Since f is increasing, we have $f(x) > L - \varepsilon$ for all $x \in (a - \delta, a) \cap D$. So, $\lim\limits_{x \to a^-} f(x)$ exists and equals L.
Similarly, we can show that $\lim\limits_{x \to a^+} f(x)$ exists and equals to $\inf\{f(x) \mid x > a\}$. $\square$

Theorem 7.10

Monotone functions can have only jump discontinuities.

Proof. Suppose a monotone function f has a discontinuity at a. Since f is monotone, both the one-sided limits at a exists (by Theorem 7.9). So, f has jump discontinuity at a. $\square$

Theorem 7.11

A strictly monotone function $f : [a, b] \to \mathbb{R}$ which has intermediate value property is continuous on $[a, b]$.

Proof. WLOG, assume f is strictly increasing. Let $x_0 \in (a, b)$ be any arbitrary point. we will show that f is continuous at x_0. By Theorem 7.9, we have

$$\lim_{x \to x_0^-} f(x) = \sup_{a < x < x_0} f(x) \leq f(x_0) \leq \inf_{x_0 < x < b} f(x) = \lim_{x \to x_0^+} f(x).$$

Let $f(x_0^-) = \sup\limits_{a < x < x_0} f(x)$ and $f(x_0^+) = \inf\limits_{x_0 < x < b} f(x)$. It suffices to show that $f(x_0^-) = f(x_0) = f(x_0^+)$ (by Theorem 7.8).
Suppose $f(x_0^-) < f(x_0)$. For any $x \in (a, x_0)$, we have $f(x) \leq f(x_0^-)$ (as f is strictly increasing). Choose any $\alpha \in (f(x_0^-), f(x_0))$. So, for any $x \in (a, x_0)$, we have $f(x) < \alpha$. But, Intermediate Value Theorem guarantees the existence of $x^* \in (x, x_0)$ such that $f(x^*) = \alpha \in (f(x), f(x_0))$ (for any $x \in (a, x_0)$), which is a contradiction! So, we must have $f(x_0^-) = f(x_0)$. Similarly, we have $f(x_0^+) = f(x_0)$. Hence, the theorem holds. $\square$

§7.7 A few standard limits

Below is a list of standard limit results you'll frequently encounter when evaluating limits of functions. You can try proving them yourself – most proofs can be found in any high school–level calculus textbook.

1. $\lim_{x\to 0} \frac{\sin x}{x} = 1$
2. $\lim_{x\to 0} \frac{\tan x}{x} = 1$
3. $\lim_{x\to 0} \frac{\ln(x+1)}{x} = 1$
4. $\lim_{x\to 0} \frac{e^x - 1}{x} = 1$
5. $\lim_{x\to 0} \frac{a^x - 1}{x} = \ln a$
6. $\lim_{x\to +\infty} \left(1 + \frac{1}{x}\right)^x = e$
7. $\lim_{x\to -\infty} \left(1 + \frac{1}{x}\right)^x = e$
8. $\lim_{x\to 0} (1+x)^{\frac{1}{x}} = e$

§7.8 More examples

Example 7.10

Assume that $\lim_{x\to a} f(x) = 1$, $\lim_{x\to a} g(x) = \infty$ and $\lim_{x\to a} g(x)\left(f(x) - 1\right) = c$. Prove that $\lim_{x\to a} f(x)^{g(x)} = e^c$.

Solution. Let $y = \lim_{x\to a} f(x)^{g(x)}$. We have

$$\ln y = \ln\left(\lim_{x\to a} f(x)^{g(x)}\right) = \lim_{x\to a}\left(\ln f(x)^{g(x)}\right)$$

as $x \mapsto \ln x$ is a continuous map. So,

$$\begin{aligned}\ln y = \lim_{x\to a}\left(\ln f(x)^{g(x)}\right) &= \lim_{x\to a}(g(x)\ln f(x)) \\ &= \lim_{x\to a} g(x)(f(x)-1)\left(\frac{\ln f(x)}{f(x)-1}\right)\end{aligned}$$

Now, as $\lim_{x\to a} f(x) = 1$, we have $\lim_{x\to a} \frac{\ln f(x)}{f(x)-1} = 1$ and it is given that $\lim_{x\to a} g(x)(f(x)-1) = c$. So, we have $\ln y = 1 \cdot c = c$. Hence, $\lim_{x\to a} f(x)^{g(x)} = y = e^c$ □

Example 7.11

What is the limit of $\left(1 + \frac{1}{n^2+n}\right)^{n^2+\sqrt{n}}$ as $n \to \infty$?

Solution. Let $y = \lim_{n\to\infty}\left(1 + \frac{1}{n^2+n}\right)^{n^2+\sqrt{n}}$. Since $x \mapsto \ln x$ is a continuous map, we have,

$$\ln y = \lim_{n\to\infty}(n^2+\sqrt{n})\ln\left(1+\frac{1}{n^2+n}\right) = \lim_{n\to\infty}\frac{n^2+\sqrt{n}}{n^2+n}\cdot\frac{\ln\left(1+\frac{1}{n^2+n}\right)}{\frac{1}{n^2+n}}$$

Now,

$$\lim_{n\to\infty} \frac{n^2+\sqrt{n}}{n^2+n} = \lim_{n\to\infty} \frac{1+\frac{1}{n^{3/2}}}{1+\frac{1}{n}} = 1.$$

Also, as $\lim_{n\to\infty} \frac{1}{n^2+n} = 0$, we have,

$$\lim_{n\to\infty} \frac{\ln\left(1+\frac{1}{n^2+n}\right)}{\frac{1}{n^2+n}} = 1.$$

So, we have, $\ln y = 1\cdot 1 = 1 \implies \lim_{n\to\infty}\left(1+\dfrac{1}{n^2+n}\right)^{n^2+\sqrt{n}} = y = e.$

Example 7.12

Suppose $f : \mathbb{R} \to \mathbb{R}$ is a function given by

$$f(x) = \begin{cases} 1 & \text{if } x = 1 \\ e^{(x^{10}-1)} + (x-1)^2 \sin\frac{1}{x-1} & \text{if } x \neq 1 \end{cases}$$

Discuss the continuity of $f(x)$ at $x = 1$.

Solution. We have

$$\lim_{x\to 1} (x-1)^2 \sin\frac{1}{x-1} = \lim_{x\to 0} y^2 \sin\frac{1}{y} = 0$$

as $\lim_{y\to 0} y = 0$ and $\lim_{y\to 0} y \sin\dfrac{1}{y} = 0$. Also, $\lim_{x\to 1}(x^{10}-1) = 0 \implies \lim_{x\to 1} e^{(x^{10}-1)} = e^0 = 1$ (as $x \mapsto e^x$ is a continuous map). So,

$$\lim_{x\to 1}\left[e^{(x^{10}-1)} + (x-1)^2 \sin\frac{1}{x-1}\right] = 1 = f(1)$$

Hence, $f(x)$ is continuous at $x = 1$.

Example 7.13

Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a monotonic function such that $\lim_{x\to\infty} \dfrac{f(2x)}{f(x)} = 1$. Show that also $\lim_{x\to\infty} \dfrac{f(cx)}{f(x)} = 1$ for each $c > 0$.

Solution. WLOG, assume f is monotonically increasing. Observe that for each $n \in \mathbb{N} \cup \{0\}$,

$$\lim_{x\to\infty} \frac{f(2^n x)}{f(x)} = \lim_{x\to\infty} \frac{f(2^n x)}{f(2^{n-1}x)} \cdot \frac{f(2^{n-1}x)}{f(2^{n-2}x)} \cdots \frac{f(2x)}{f(x)} = 1,$$

as $\lim\limits_{x\to\infty} \dfrac{f(2^k x)}{f(2^{k-1}x)} = 1$ for each $k \geq 1$. Now, for each $c \geq 1$, there exists $N \in \mathbb{N} \cup \{0\}$ such that $2^N \leq c < 2^{N+1}$. Since f is monotonically increasing, we have,

$$\frac{f(2^N x)}{f(x)} \leq \frac{f(cx)}{f(x)} \leq \frac{f(2^{n+1}x)}{f(x)}.$$

By using Squeeze Theorem, we obtain that $\lim\limits_{x\to\infty} \dfrac{f(cx)}{f(x)} = 1$ for each $c \geq 1$.

Now for $0 < c < 1$, let $c' = \frac{1}{c} > 1$. So, $\lim\limits_{x\to\infty} \dfrac{f(c'x)}{f(x)} = 1 \implies \lim\limits_{x\to\infty} \dfrac{f(\frac{1}{c}x)}{f(x)} = 1 \implies \lim\limits_{t\to\infty} \dfrac{f(t)}{f(ct)} = 1 \implies \lim\limits_{x\to\infty} \dfrac{f(ct)}{f(t)} = 1$. So, we have $\lim\limits_{x\to\infty} \dfrac{f(cx)}{f(x)} = 1$ for all $c > 0$. □

Example 7.14

Let $f : [0,1] \to [-1,1]$ be a function satisfying $f(2x) = 3f(x)$ for all $x \in \left[0, \frac{1}{2}\right]$. Show that $\lim\limits_{x\to 0^+} f(x) = 0$.

Solution. Putting $x = 0$ in the given equation, we get $f(0) = 0$. So, we have to show $\lim\limits_{x\to 0^+} f(x) = 0$. Observe that the following holds:

$$\begin{aligned}
f(x) &= \frac{f(2x)}{3} \qquad \forall\, x \in \left[0, \frac{1}{2}\right] \\
f(x) &= \frac{f(2^2x)}{3^2} \qquad \forall\, x \in \left[0, \frac{1}{2^2}\right] \\
f(x) &= \frac{f(2^3x)}{3^3} \qquad \forall\, x \in \left[0, \frac{1}{2^3}\right] \\
&\vdots \\
f(x) &= \frac{f(2^nx)}{3^n} \qquad \forall\, x \in \left[0, \frac{1}{2^n}\right], \quad n \in \mathbb{N}
\end{aligned}$$

For each $x \in [0,1]$, we have $|f(x)| \leq 1$. Now, fix any $\varepsilon > 0$. There exists $N \in \mathbb{N}$ such that $\frac{1}{3^N} < \varepsilon$ (for instance, just take $N = \lceil \log_3(1/\varepsilon) \rceil$). Let $\delta = \frac{1}{2^N}$. Then $\forall\, x \in (0, \delta) \subset \left[0, \frac{1}{2^N}\right]$, we have

$$|f(x)| = \frac{|f(2^N x)|}{3^N} \leq \frac{1}{3^N} < \varepsilon.$$

Hence, we have $\lim\limits_{x\to 0^+} f(x) = 0 = f(0)$. □

Example 7.15

Suppose a function $f : (-a, a) \setminus \{0\} \to (0, +\infty)$ satisfies $\lim\limits_{x \to 0} \left(f(x) + \dfrac{1}{f(x)} \right) = 2$. Show that $\lim\limits_{x \to 0} f(x) = 1$.

Solution. Observe that

$$f(x) + \frac{1}{f(x)} - 2 = (f(x) - 1)\left(1 - \frac{1}{f(x)}\right).$$

Note that $f(x) + \frac{1}{f(x)} \geq 2$. So, by hypothesis, given $\varepsilon > 0$, there exists $\delta > 0$ such that

$$0 \leq f(x) + \frac{1}{f(x)} - 2 < \varepsilon \quad \forall\, x \in (-\delta, \delta) \setminus \{0\}.$$

So, we have,

$$\begin{aligned}(f(x) - 1)^2 &\leq (f(x) - 1)^2 + \left(\frac{1}{f(x)} - 1\right)^2 \\ &= \left(f(x) - 1 + \frac{1}{f(x)} - 1\right)^2 + 2(f(x) - 1)\left(1 - \frac{1}{f(x)}\right) < \varepsilon^2 + 2\varepsilon\end{aligned}$$

$$\implies |f(x) - 1| < \sqrt{\varepsilon^2 + 2\varepsilon}$$

Therefore, $\lim\limits_{x \to 0} f(x) = 1$. □

§7.9 Exercise 6

Problem 7.1. Show that the two definitions of limit (i.e., Definition 6.3 and Definition 6.4) are equivalent.

Problem 7.2. Prove Theorem 6.4 (Using the sequential definition should make things obvious, because you can invoke the usual limit theorems for sequences).

Problem 7.3. Consider the function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1 & \text{if } x \in \mathbb{Q} \\ 0 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Show that $\lim\limits_{x \to a} f(x)$ does not exist for any $a \in \mathbb{R}$.

Problem 7.4. Calculate the following limits:

1. $\lim_{x\to 0} x^2 \left(1+2+3+\cdots+\left\lfloor \frac{1}{|x|} \right\rfloor\right)$

2. $\lim_{x\to 0^+} x\left(\left\lfloor \frac{1}{x} \right\rfloor + \left\lfloor \frac{2}{x} \right\rfloor + \cdots + \left\lfloor \frac{k}{x} \right\rfloor\right), \quad k \in \mathbb{N}.$

Problem 7.5. Find the following limits:

1. $\lim_{x\to 0} \frac{x}{x+\sin x}$

2. $\lim_{x\to 0} \frac{x^4+x^6}{e^x-1+x^2}$

3. $\lim_{x\to\infty} \frac{p(x)}{e^x}$, where p is a polynomial

4. $\lim_{x\to\infty} \sqrt{x}(\sqrt{x+4}-\sqrt{x})$

5. $\lim_{x\to 0} \frac{\cos(\frac{\pi}{2}\cos x)}{\sin(\sin x)}$

6. $\lim_{n\to\infty} \tan^n\left(\frac{\pi}{4}+\frac{1}{n}\right), \quad n \geq 1$

If you want to practice more on calculating the limit of different functions, you can try the problems from here and here.

Problem 7.6. Determine, with proof, the value of

$$\lim_{x\to 0} \frac{1-\cos(x)\cos(2x)\cdots\cos(nx)}{x^2}.$$

Problem 7.7. Determine, with proof, the value of the limit $\lim_{x\to\infty}\left(1+\frac{1}{x^2+\cos x}\right)^{x^2+x}$

Problem 7.8. Let $f:(0,\infty)\to(0,\infty)$ be a function satisfying $\lim_{x\to 0}\frac{f(3+x)-f(3)}{x} = \frac{f(3)}{3} > 0$. Then find the limit

$$\lim_{x\to\infty}\left(\frac{f\left(3+\frac{3}{x}\right)}{f(3)}\right)^x.$$

Problem 7.9. Suppose $\lim_{x\to a}\frac{f(x)-f(a)}{x-a} = p$. Then find the following limit in terms of p:

$$\lim_{n\to\infty} n\left(f\left(a+\frac{1}{a}\right)+f\left(a+\frac{2}{a}\right)+\cdots+f\left(a+\frac{k}{a}\right)-kf(a)\right), \quad k \in \mathbb{N}.$$

Problem 7.10. Assume f is defined on a deleted neighborhood of a and

$$\lim_{x\to a}\left(f(x)+\frac{1}{|f(x)|}\right) = 0.$$

Determine $\lim_{x\to a} f(x)$.

Problem 7.11. Prove that if f is a bounded function on $[0, 1]$ satisfying $f(ax) = bf(x)$ for $0 \le x \le \frac{1}{a}$ and $a, b > 1$, then $\lim_{x \to 0^+} f(x) = f(0)$.

Problem 7.12. If f defined on (a, ∞) be bounded on each finite interval (a, b). If for a non-negative integer k, $\lim_{x\to\infty} \dfrac{f(x+1) - f(x)}{x^k}$ exists, then show that

$$\lim_{x\to\infty} \frac{f(x)}{x^{k+1}} = \frac{1}{k+1} \lim_{x\to\infty} \frac{f(x+1) - f(x)}{x^k}.$$

***Problem 7.13.** Let $f : (0, 1) \to \mathbb{R}$ be a function satisfying $\lim_{x\to 0^+} f(x) = 0$ and such that there exists $\lambda \in (0, 1)$ such that $\lim_{x\to 0^+} \dfrac{f(x) - f(\lambda x)}{x} = 0$. Prove that $\lim_{x\to 0^+} \dfrac{f(x)}{x} = 0$.

***Problem 7.14** (Romania district 2025)**.** Let $f : [0, \infty) \to [0, \infty)$ be a continuous and bijective function, such that

$$\lim_{x\to\infty} \frac{f^{-1}(f(x)/x)}{x} = 1.$$

Show that $\lim_{x\to\infty} \dfrac{f(x)}{x} = \infty$ and $\lim_{x\to\infty} \dfrac{f^{-1}(ax)}{f^{-1}(x)} = 1$ for any $a > 0$.

§8 Miscellaneous exercise

Problem 8.1. Let $f : [a, b] \to \mathbb{R}$ be a continuous function and $\{a_n\}_{n\ge 1} \subset [a, b]$ be a sequence of reals. Prove that there exists $x \in [a, b]$ such that

$$f(x) = \sum_{n=1}^{\infty} \frac{f(a_n)}{2^n}$$

Problem 8.2. Suppose that $f : [1, 2] \to \mathbb{R}$ is a continuous function that satisfies

$$f(x) = \sum_{n=1}^{\infty} \frac{f(x^{1/n})}{2^n}$$

for every $x \in [1, 2]$. Show that f must be a constant function.

Problem 8.3. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function satisfying $f(x) = f(x^2)$ for all $x \in \mathbb{R}$. Prove that f is constant.

Problem 8.4. Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that

$$f\left(r + \frac{1}{n}\right) = f(r) \quad \forall\, r \in \mathbb{Q} \text{ and } n \in \mathbb{N},$$

Prove that f is constant.

Problem 8.5. Suppose $f : \mathbb{R} \to \mathbb{R}$ is continuous on $\mathbb{R}$ and satisfies $f(x + y) = f(x) + f(y)\ \forall\ x, y \in \mathbb{R}$. Show that $f(x) = cx\ \forall\ x \in \mathbb{R}$, where $c = f(1)$.

Problem 8.6. Suppose $f : \mathbb{R} \to \mathbb{R}$ be a function continuous at 0 and satisfies $f(x+y) = f(x) + f(y)\ \forall x, y \in \mathbb{R}$. Show that f is continuous on $\mathbb{R}$ and hence conclude that $f(x) = cx\ \forall\ x \in \mathbb{R}$, where $c = f(1)$.

Problem 8.7. A real-valued function f defined in (a, b) is said to be *convex* if

$$f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$$

whenever $a < x, y < b$ and $0 < \lambda < 1$. Prove that every convex function is continuous.

Problem 8.8 (RNMO 1979)**.** Prove that there exists no continuous function $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) \in \mathbb{Q} \iff f(x + 1) \notin \mathbb{Q}.$$

Problem 8.9 (Simon Marais 2024 B2)**.** Determine all continuous functions $f : \mathbb{R} \setminus \{1\} \to \mathbb{R}$ that satisfy

$$f(x) = (x + 1)f(x^2),$$

for all $x \in \mathbb{R} \setminus \{1\}$.

Problem 8.10 (IMC 2008)**.** Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ such that

$$f(x) - f(y) \in \mathbb{Q} \quad \forall\ x - y \in \mathbb{Q}.$$

Problem 8.11 (Romania district 2023)**.** Let $f : [a, b] \to [a, b]$ be a continuous function, it is known that there exist $\alpha, \beta \in (a, b)$ such that $f(\alpha) = a$ and $f(\beta) = b$. Prove that the function $f \circ f$ has at least three fixed points.

Problem 8.12 (RNMO 2019)**.** Let $f : [0, \infty) \to \mathbb{R}$ be a continuous function, constant on $\mathbb{Z}_{\geq 0}$. For any $0 \leq a < b < c < d$ which satisfy $f(a) = f(c)$ and $f(b) = f(d)$ we also have $f(\frac{a+b}{2}) = f(\frac{c+d}{2})$. Prove that f is constant on $[0, \infty)$.

Problem 8.13. Let $f, g : \mathbb{R} \to \mathbb{R}$ be two functions such that $f \circ g$ is continuous on $\mathbb{R}$ and f is a strictly increasing function. Show that g is continuous on $\mathbb{R}$.

Problem 8.14 (SEEMOUS 2023)**.** Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous, strictly decreasing function such that $f([0, 1]) \subseteq [0, 1]$. For each positive integer, prove that there exists a unique $a_n \in (0, 1)$, solution of the equation $f(x) = x^n$. Moreover, if $\{a_n\}$ is the sequence defined as above, prove that $\lim\limits_{n\to\infty} a_n = 1$

Problem 8.15 (Putnam 1991)**.** Let $a, b \in \left(0, \frac{1}{2}\right)$ and let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function such that

$$f(f(x)) = af(x) + bx\ \ \forall\ x \in \mathbb{R}.$$

Prove that $f(0) = 0$

Problem 8.16. Find all continuous functions $f : \mathbb{R} \to \mathbb{R}$ satisfying $f(0) = 1$ and $f(2x) - f(x) = x$, for all $x \in \mathbb{R}$.

Problem 8.17. A function $f : [0,1] \to \mathbb{R}$ satisfies $f(0) < 0$ and $f(1) > 0$ and there exists a function g continuous on $[0,1]$ and such that $f + g$ is decreasing. Prove that the equation $f(x) = 0$ has a solution in the open interval $(0,1)$.

Problem 8.18. Let $f, g : [0,1] \to [0,1]$ be two functions such that g is monotonic, surjective and $|f(x) - f(y)| \leq |g(x) - g(y)|$, for any $x, y \in [0,1]$. Prove that f is continuous and there exists some $x_0 \in [0,1]$ with $f(x_0) = g(x_0)$.